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SariSite: Enhancing Accessibility of Localized Sari-Sari Stores in Philippine Subdivision Through Online Ordering

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Chapter I

Introduction

Sari-sari stores are an integral part of Filipino communities, serving as convenient sources of daily necessities within neighborhoods. However, many of these stores remain limited to traditional face-to-face transactions, which can be time-consuming and less accessible to residents with busy schedules or mobility constraints. With the increasing reliance on digital platforms for everyday needs, integrating an online ordering system for localized sari-sari stores offers a practical solution to improve accessibility and convenience. This study explores how such a system can enhance customer experience, streamline store operations, and strengthen the role of sari-sari stores in Philippine subdivisions.

1.1 Background of the Study

Even small neighborhood sari-sari stores are now run with the help of technology. However, the majority of stores are still open spaces, humans walk in and buy things, and purchase face-to-face (personally with staff) more often than not, which can be slow and cumbersome. The majority of people use the internet daily, so having access to an online store where customers can order products such as sari-sari items will help reach more customers faster, and clients can purchase without leaving their homes. The study centers on creating a sari-sari store online ordering and browsing system to the website of a sari-sari store inside the subdivision.

1.2 Statement of the Problem

Despite being vital to neighborhood life, most sari-sari stores in Philippine subdivisions still rely on traditional, walk-in purchasing methods. This setup often limits accessibility for residents who may face time constraints, distance, or mobility challenges. While digital platforms are increasingly used in various retail sectors, sari-sari stores have yet to fully adapt to these technologies. The lack of an online browsing and ordering system creates a gap in convenience and efficiency, which this study seeks to address by exploring

how digital integration can enhance the accessibility and service of localized sari-sari stores.

This study aims to answer the following questions:

1. How can a website with an online browsing and ordering system improve the sari-sari store's sales and customer service?
2. What features should the website include to be user-friendly for both customers and the store owner, including order delivery options?
3. Will customers prefer ordering online with delivery rather than going to the store physically?
4. What challenges might the store owner face in managing orders and deliveries through the website?

1.3 Objectives of the Study

This study was conducted to improve the accessibility and efficiency of localized sari-sari stores in Philippine subdivisions through the development of an online browsing and ordering system. To clearly define the scope of the research, the objectives are divided into general and specific objectives.

1.3.1 General Objective

The general objective of this study is to develop an online browsing and ordering system that enhances the accessibility and efficiency of localized sari-sari stores within Philippine subdivisions.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. Design a simple and easy-to-use website for regular customers to browse and order available products online.
2. Make the website accessible for all users with more features being available to users with accounts such as a history of their past purchases, notifications for when an item gets restocked, and the delivery service.

3. Add a system that informs customers if the delivery service is currently available or not.
4. Provide the store owner with an efficient system to manage online orders and deliveries.
5. Evaluate if the website improves customer satisfaction and increases sales.
6. Identify any problems or limitations in using the website's ordering and delivery features.

1.4 ISO Quality and Acceptance Criteria

To ensure that the developed web application meets international software quality standards, the project will be evaluated based on the **ISO/IEC 25010 Software Quality Model**. The system will be assessed using the following quality characteristics and acceptance criteria:

- **Functionality** – The application must perform all essential tasks accurately, such as customer registration, product browsing, order placement, and payment transactions. Each function should operate as intended without major errors or inconsistencies.
- **Usability** – The interface should be intuitive and user-friendly, allowing both customers and store owners to easily navigate through pages, access features, and complete actions with minimal guidance.
- **Reliability** – The system must demonstrate consistent and stable performance even during extended use, ensuring that no data is lost or corrupted due to unexpected failures or interruptions.
- **Performance Efficiency** – The application should load pages and process transactions quickly, maintaining smooth responsiveness even when multiple users are accessing the system simultaneously.
- **Security** – All sensitive information, such as login credentials, payment details, and personal data, must be securely stored and protected using authentication and encryption methods to prevent unauthorized access.
- **Compatibility** – The system should function correctly across various web browsers, ensuring accessibility and consistency for all users regardless of platform.

- **Maintainability** – The project's source code should be organized and modular, allowing easy debugging, updates, and future feature integration without disrupting system operations.
- **Portability** – The application should be easily deployable and operable on different platforms, including web browsers and Android devices, without requiring major code modifications.

1.5 Significance of the Study

This study is significant because it addresses the modernization of localized sari-sari stores, which play an essential role in the daily lives of Filipino communities. By introducing an online browsing and ordering system, the research contributes not only to technological innovation but also to community development, convenience, and business sustainability. The following groups stand to benefit from the outcomes of this study:

Customers. For customers, the proposed system offers a more convenient and efficient way of purchasing goods from sari-sari stores. Instead of being limited to physical visits, customers can browse available products, place orders, and request deliveries through the website application. This will particularly benefit individuals with busy schedules, mobility challenges, or those who simply prefer the convenience of digital transactions. By improving accessibility, the website application allows customers to save time, avoid long queues, and enjoy a more user-friendly purchasing experience that aligns with modern lifestyles.

Sari-sari Store Owners. Sari-sari store owners will benefit from the system by gaining a platform that extends their business operations into the digital space. With the website application, they can reach more customers within the subdivision and improve their sales performance through online visibility and accessibility. The system can also provide features that help streamline store management, such as monitoring inventory, tracking orders, and analyzing customer preferences. Furthermore, it enables store owners to adapt to changing consumer behaviors, remain competitive in a technology-driven marketplace, and potentially increase profitability while reducing inefficiencies in traditional operations.

Future Researchers. This study can serve as a useful reference for future researchers who wish to explore the integration of technology into small-scale businesses. The findings and outcomes may provide a foundation for further innovations in e-commerce solutions for microenterprises, especially in localized settings. It can also encourage future studies on how community-based businesses can adopt digital transformation strategies to remain relevant and sustainable in an increasingly digital economy. By building upon this study, future researchers can contribute to the continuous improvement of small business technologies in the Philippines and beyond.

1.6 Scope and Limitations

This project focuses on developing an online and delivery system for sari-sari stores located in subdivisions. It aims to create a user-friendly interface for customers and store owners, allowing product browsing, ordering, and purchasing. The system displays product listings and delivery availability, helping owners manage orders efficiently. It uses cloud computing for accessibility and supports two payment methods—GCash and Cash—for convenience, along with live chat between customers and owners. However, its coverage is limited to subdivisions, inventory updates are manual, and delivery depends on the owner's available resources. Lastly, the project can only be used by one sari-sari store owner.

Chapter II

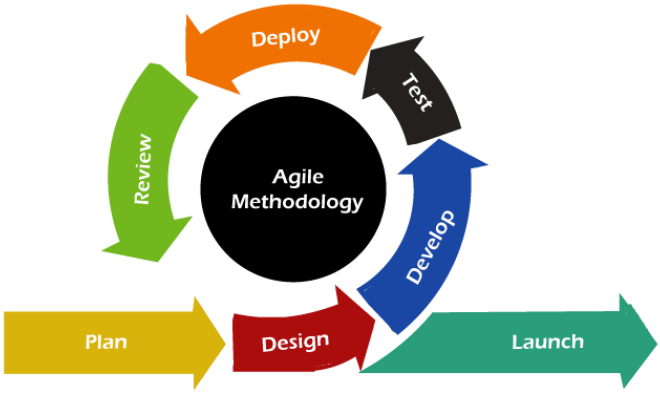
Methodology of the Study

This chapter explains the methodology employed in the creation of the proposed online delivery and ordering system for a sari-sari store in a subdivision. For the achievement of the objectives described in Chapter 1, researchers employed the Agile Software Development Model. Agile is adaptive, iterative, and feedback-driven, which makes it suited for small-scale projects where user experience and ongoing improvement are paramount. Employing this strategy, the researchers made sure that the system would be functional, user-friendly, and responsive to the requirements of customers as well as the store owner.

2.1 Agile Software Development Model

The **Agile Model** focuses on incremental progress through short development cycles, known as **sprints**. Each sprint produces a working version of the system that can be tested, evaluated, and improved. Unlike traditional methods, Agile encourages flexibility, collaboration, and frequent feedback. This is essential for the sari-sari store project, as the system must adapt to user expectations while maintaining simplicity and efficiency.

2.2 Agile Process Phases



2.2.1 Planning

The planning stage involved setting the project goals and identifying the system requirements. The researchers determined that the core features must include:

- A customer login system using Firebase Authentication (Google Sign-In) for secure user access.
- An online ordering and cart system for customers.
- Payment options via GCash and Cash.
- A store owner dashboard for managing orders and product listings.
- A delivery feature that allows customers to place orders for home delivery.

The interface was planned to remain simple and mobile-friendly to cater to a wide range of users.

2.2.2 Requirements Analysis and Design

In this phase, the system's structure and features were analyzed and outlined. Key components included:

- Customer Interface: browsing products, adding items to the cart, placing orders, and logging in via Firebase Authentication (Google Sign-In).
- Delivery Management: displaying availability and updating customers on order status.
-
- Store Owner Dashboard: allowing the owner to manage product listings and track orders.

For the system's design, **Flutter (built with Dart)** was selected to create a modern, cross-platform interface that is intuitive and visually appealing. **MongoDB** served as the main backend database for storing product, order, and user information. Wireframes were drafted to ensure that navigation remained smooth and user-friendly.

The backend services were developed using **Node.js and Express.js**, enabling efficient data processing, secure communication with the MongoDB database, and smooth integration with the Flutter front-end interface.

2.2.3 Iteration / Development (Sprints)

Development was carried out in iterative sprints. Each sprint focused on specific features, for example:

- Sprint 1: Customer login using Firebase Authentication (Google Sign-In).
- Sprint 2: Product catalog and cart system connected to MongoDB..
- Sprint 3: Store owner dashboard and management tools.
- Sprint 4: Delivery feature with the Flutter interface.

At the end of each sprint, a working version of the system was produced and tested using the Flutter front-end and Node.js/Express backend connected to MongoDB.

2.2.4 Testing

Testing was conducted after each sprint to confirm that the features functioned properly. The types of testing included:

- **Functional Testing:** ensuring Firebase login and MongoDB database connections, product ordering, and delivery options worked as intended.
- **API Testing:** verified backend routes and database connections using Bruno to ensure data consistency and system reliability. **Usability Testing:** evaluating if the Flutter-built interface was simple and easy to navigate.
- **Compatibility Testing:** checking that the system performed well on multiple devices and browsers.

Any issues found during testing were logged and corrected before the next sprint.

2.2.5 Review and Feedback

At the end of each sprint, feedback was collected from potential users and the store owner. Their input guided further development, ensuring that improvements were made gradually and aligned with real user needs.

2.2.6 Deployment

Once the system reached a functional state, it was deployed using a Node.js and Express.js backend integrated with a Flutter front-end application and a MongoDB database. The system is hosted on a compatible web deployment platform, making it accessible to residents within the subdivision. It integrates Firebase Authentication for login and supports GCash and Cash payment methods.

2.2.7 Maintenance and Continuous Improvement

After deployment, the system entered the maintenance phase. This stage includes monitoring performance, updating product lists, fixing bugs, and adding new features when necessary. Continuous improvement ensures that the sari-sari store system remains useful, reliable, and secure for long-term use.

2.2.8 Development Tools and Environment

The following tools and platforms were used in the development of the system:

- **Programming Languages and Frameworks:** Dart and Flutter (for cross-platform frontend development), Node.js and Express.js (for backend API and server-side operations)
- **Database and Hosting:** MongoDB (for data storage and backend management)
- **Authentication:** Firebase Authentication (Google Sign-In) for secure login and user identity management
- **API Testing:** Bruno (for testing and validating backend routes and data flow)
- **Version Control:** GitHub (for collaborative development and version tracking)
- **Development Environment:** Visual Studio Code (for integrated development and debugging)

These tools were selected for their flexibility, scalability, and suitability for full-stack application development. Flutter ensures a responsive and intuitive interface, while Node.js, Express, and MongoDB handle backend operations efficiently. Google Cloud Authentication provides secure login and account management, and Bruno assists in ensuring reliable API communication throughout the system.

2.3 Summary

In conclusion, the researchers adopted the Agile Software Development Model to guide the development of the sari-sari store online browsing and ordering system. The iterative process of planning, design, development, testing, and feedback allowed the researchers to build a functional system that meets the objectives of the study.

By integrating Google Cloud Authentication, MongoDB, Node.js, Express.js, and Flutter/Dart, along with GitHub for version control, the researchers ensured that the project remains secure, collaborative, and user-friendly. This methodology not only supports the technical requirements of the study but also provides a foundation for future enhancements.

Chapter III

Data Gathering Procedures and Outputs

This chapter presents the procedures undertaken to gather data in order to determine the system requirements for the proposed Online Browsing and Ordering System for localized sari-sari stores within Philippine subdivisions. The primary objective of this stage was to identify the existing challenges faced by customers and store owners, as well as their expectations for an online platform that integrates ordering and delivery features.

The findings from this data gathering activity served as the foundation for the design and development of the system described in Chapter II. To achieve this, the researchers utilized surveys and interviews with selected respondents to obtain relevant information regarding current purchasing practices, accessibility concerns, and the preferred functionalities of the proposed system.

3.1 Stakeholders / Respondents

Respondents were carefully selected based on their active involvement in sari-sari store operations and transactions. Their input was essential in capturing both the customer perspective and the operational requirements of store owners.

Role	Number of Respondents	Rationale for Selection
Sari-sari Store Owners	3	Provide insights into inventory management, sales tracking, delivery challenges, and system requirements for business operations.

Customers (Subdivision Residents)	30	Regularly purchase goods from sari-sari stores; provide feedback on accessibility, delivery preferences, and usability of an online system.

3.2 Data Collection Methods

To obtain comprehensive and reliable data, the researchers employed an online survey method as well as an Interview with one of the sari-sari store owners. Two separate questionnaires were distributed — one for customers, one for sari-sari store owners — each tailored to their role in the study.

The customer questionnaire primarily used a Likert-scale format to generate quantitative data on purchasing behaviors, delivery preferences, and satisfaction with existing sari-sari store services. Meanwhile, the questionnaires for sari-sari store owners contained open-ended questions to obtain qualitative insights on operational challenges and technical requirements relevant to the development of the proposed online sari-sari store system.

3.3 Instruments / Tools

Data regarding system requirements were also gathered through a one-on-one structured interview with a key stakeholder. The interview focused on identifying the specific features, functions, and expectations needed for the software to be developed. Through direct discussion, the researchers obtained detailed insights and clarifications that could not be captured through the questionnaire alone. This single interview provided valuable information about user needs, operational processes, and essential technical considerations that would guide the system's design and development.

Besides that, another instrument utilized in the study was an online questionnaire developed through Google Forms. The platform was chosen for its accessibility, ease of distribution, and ability to automatically record responses. The questionnaire was specifically designed to gather information about customer behavior and preferences

related to sari-sari store transactions. It included questions about purchasing frequency, interest in online ordering, willingness to use delivery services, confidence in using digital platforms, and the perceived usefulness of features such as online browsing, and pre-order options.

Likert Scale Interpretation - To measure the level of agreement among the respondents, a 4-point Likert scale was used. The scale helped determine how strongly customers agreed or disagreed with each statement related to the online sari-sari store system.

Scale	Description	Interpretation
4	Strongly Agree	The respondents fully agree and find the feature highly beneficial.
3	Agree	The respondents generally agree and consider the feature useful.
2	Disagree	The respondents slightly disagree and find the features less necessary.
1	Strongly Disagree	The respondents do not agree and find no relevance in the future.

A. Survey for Customers

A 9-item form designed to assess customer satisfaction, ordering habits, and feature preferences such as delivery options, and payment methods, using a 4-point Likert scale (e.g., 1 = Strongly Disagree to 4 = Strongly Agree) to measure the level of agreement or satisfaction for each item. One question is open-ended to allow respondents to elaborate on their experiences and provide qualitative insights beyond the Likert scale responses.

- Do you often purchase from sari-sari stores in your subdivision? Why or why not?
- Would you agree that ordering from a sari-sari store online instead of visiting physically would be more convenient?

- If delivery service were available, how likely are you to use it when buying daily necessities?
- Do you believe that having an online browsing and ordering system will improve your shopping experience with sari-sari stores?
- Would seeing the available stock of items online make it more convenient for you to shop at a sari-sari store?
- How confident are you in using a website or app for ordering from a sari-sari store?
- How important is it for you to receive notifications (e.g., order confirmation, delivery status, item restocked) from the sari-sari store's online system?
- I would find a pre-order or reservation feature useful for items that run out quickly.
- Would you recommend an online sari-sari store to your neighbors or family if it is easy to use and makes buying from sari-sari stores more convenient?

B. One-on-One Interview with Main Stakeholder

A structured one-on-one interview was conducted with the main stakeholder, who is also a sari-sari store owner, to obtain detailed insights into business operations, user needs, and the specific system requirements needed for the proposed software.

Questions Regarding Business Background

- How long have you owned your sari-sari store?
- Who usually manages the store besides you?
- How many people help you run the store?
- What are your store's operating hours?
- What types of products do you mainly sell?

Business Operations and Workflow

- Can you describe your step-by-step process from receiving an order to completing the transaction?
- How do you handle situations when an item becomes out of stock?
- What factors typically slow down your operations or cause delays?
- What are the peak times for your store, and how do you manage workload during these hours?

Policies and Procedures

- Do you have existing policies regarding order cancellations, returns, or replacements?
- How do you verify customer orders before preparing or releasing items?
- Do you have guidelines for pricing, discounts, or bundled items that the system should follow?
- How often do you conduct inventory checks, and what procedures do you follow to ensure accuracy?
- Do you have rules or standards for delivery services (minimum order amount, delivery areas, delivery fees)?

User Needs and Expectations

- Which daily tasks would you prefer to automate through the system?
- How important is mobile accessibility for you when using the system?
- Would you prefer a simple interface or a more advanced system with analytics?

Technical Feasibility and System Design

- Do you have access to a stable internet and a device suitable for using an online system?
- What concerns do you have regarding the privacy and security of customer or store data?
- How confident are you in using digital tools, and would you require orientation or training?
- How often would you need to generate sales, expense, or inventory reports?
- Are there any other tools or integrations that would be useful (e-receipts, payment methods, delivery tracking, etc.)?

C. Survey for Sari-Sari Store Owners

The structured questionnaire containing open-ended questions was distributed to other sari-sari store owners besides the one that was interviewed in order to gather insights into business operations and system requirements. The questions included:

- What challenges do you encounter in managing daily transactions?
- How do you currently monitor and record your inventory and sales?
- What difficulties do you experience when handling customer orders and deliveries?
- What features would you find helpful in an online sari-sari store management system?
- What potential challenges do you anticipate in adopting an online system for your store?

3.4 Data Analysis Plan

The gathered data were analyzed systematically through the following steps:

- **Categorization** — Customer survey responses were grouped according to common themes such as convenience, delivery, online browsing, usability, and pre-ordering. Responses from the open-ended questionnaires of sari-sari store owners were categorized based on operational challenges, desired system features, and potential adoption barriers.
- **Quantitative Analysis (Weighted Mean)** — Likert-scale responses from customers were assigned numerical values and analyzed using weighted mean to determine the level of agreement for each statement. This helped identify features most valued by customers.
- **Prioritization** — Store owner responses regarding desired system features were classified as “must-have,” “nice-to-have,” or “optional” based on frequency and importance mentioned by respondents. Customer survey items were interpreted according to the weighted mean results.
- **Summarization** — Key results were organized in tables to highlight recurring issues, preferences, and suggested solutions for both customer and store owner data.
- **Validation** — Customer survey results were cross-checked with store owner questionnaire responses to ensure consistency and reliability of the findings, guiding the definition of system requirements.

3.5 Summary of Customer Survey Results

Statement	Weighted Mean	Interpretation
Ordering from a sari-sari store online instead of visiting physically would be more convenient.	3.83	Strongly Agree
If delivery service were available, I would likely use it when buying daily necessities.	3.70	Strongly Agree
Having an online browsing and ordering system will improve my shopping experience.	3.75	Strongly Agree
Seeing available stock of items online makes shopping more convenient.	3.68	Strongly Agree
I am confident in using a website or app for ordering from a sari-sari store.	3.52	Strongly Agree
A pre-order or reservation feature for sold-out items would be useful.	3.63	Strongly Agree
I would recommend an online sari-sari store to my neighbors or family.	3.77	Strongly Agree
Overall Weighted Mean	**3.68**	**Strongly Agree**

Interpretation:

The overall weighted mean of 3.68 indicates that customers **strongly agree** with the implementation of an online sari-sari store system. Most respondents believe that online ordering and delivery would **increase convenience**, **save time**, and **improve accessibility**. These findings support the system's main goal of enhancing customer satisfaction and modernizing local store transactions within the subdivision.

3.6 Sari-Sari Store Owner Survey Results

The responses from sari-sari store owners provided qualitative insights into their current business operations, challenges, and their views on adopting an online sari-sari store management system. Their feedback helped the researchers identify the necessary features and possible barriers in implementing the proposed system.

Challenges Encountered in Managing Daily Transactions

Common Responses	**Summary Interpretation**
Manual recording of daily sales and stock updates	Store owners rely on notebooks, which makes tracking time-consuming and prone to errors.
Difficulty monitoring inventory levels	Stocks are only checked when customers inquire or when items run out.
Managing multiple customer orders	Handling many customers at once can lead to confusion and incomplete records.

Interpretation:

The findings show that sari-sari store owners face difficulties in manually recording sales and monitoring inventory. These limitations highlight the need for an automated system that organizes and tracks sales transactions efficiently.

Desired Features in an Online Store Management System

Suggested Features	**Purpose / Benefit**
Inventory tracking and automatic stock updates	Allows real-time monitoring of product availability.
Order and delivery tracking	Helps manage pending and completed customer deliveries.
Sales summary and reporting tool	Simplifies monitoring of income and product performance.

Interpretation:

Store owners prefer features that automate daily tasks and provide better control over sales and deliveries. They want tools that are simple, reliable, and easy to use.

Potential Challenges in Adopting an Online System

Identified Challenge	**Explanation**
Limited technical knowledge	Some owners are not yet familiar with using mobile sites or online systems.
Dependence on internet connection	Online systems may not function proper
Adjustment period and maintenance	Owners may need time to adapt and maintain updated product data.

Interpretation:

While store owners are open to adopting digital systems, they anticipate challenges related to training and connectivity. Proper guidance and user-friendly design will help address these barriers.

3.7 Summary of Store Owner Survey Results

Overall, sari-sari store owners expressed interest in adopting an online system if it helps simplify their operations. They view automation as a potential solution to reduce manual work and improve order tracking, but they also require a system that is easy to learn and maintain.

3.8 Output: Defined Software Requirements

3.8.1 Functional Requirements

- Customer login and authentication via Firebase Authentication (Google Sign-In).
- Product catalog with browsing, categories, and cart system.

- Delivery service integration with status tracking and availability indicator.
- Payment options supporting both GCash and Cash.
- Store owner dashboard for order monitoring, product management, and sales tracking.

3.8.2 Non-Functional Requirements

- Mobile-friendly and responsive interface designed with Flutter and Dart.
- Accessibility through any modern web browser.
- Secure authentication and data handling via MongoDB.
- Fast response time for browsing, checkout, and delivery tracking.
- Order and delivery for customers.

3.8.3 Data Requirements

- **Product:** id, name, description, price, stock quantity, category.
- **User (Customer):** id, name, email, address, phone number.
- **Order:** id, selected items, total amount, payment method, status, date/time.
- **Delivery:** id, assigned personnel (if applicable), delivery status, timestamp.

3.8.4 User Interface / Experience Requirements

- Intuitive and simple layout designed for all age groups.
- Clear product listings with images, prices, and availability.
- Search and filter functions for faster navigation.
- Order history for registered customers.
- Dashboard with order summaries and sales reports for store owners.

3.8.5 Constraints / Assumptions / Dependencies

- Service coverage limited to the subdivision only.
- Inventory updates will remain manual for store owners.
- Internet connection is required for both customers and store operators.
- Initial payment options limited to GCash and Cash.

3.9 Summary / Conclusion

This chapter presented the procedures employed in gathering data and the corresponding outputs used to define the requirements of the proposed system. Through surveys and interviews, the researchers identified customer preferences for convenience, delivery options, and accessible payment methods, as well as the challenges faced by store owners in managing walk-in transactions and inventory.

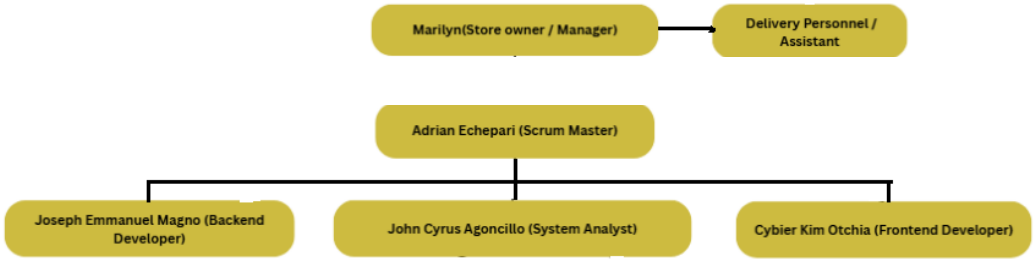
The findings provided the basis for establishing the system's functional and non-functional requirements, which include features such as online ordering, delivery tracking, customer authentication, and a management dashboard for store owners. These requirements will serve as the framework for the design and development of the Online Browsing and Ordering System for sari-sari stores, which is further elaborated in the succeeding chapter.

Chapter IV

Documentation of the Current System

4.1 Organizational Chart

The organizational chart below illustrates the structure of Alyn Store and the SariSite development team. It highlights the individuals directly involved in the store's daily operations and the project's system development.



4.2 Roles and Responsibilities of Personnel Involved in SariSite Development

Name and Position	Duties and Responsibilities	Relation to the Study
Marilyn (Owner / Manager)	Manages daily store operations, handles customer payments, monitors product inventory, and records transactions manually.	Serves as the primary administrator of the SariSite system, managing orders, inventory, and delivery requests.
Delivery Personnel / Assistant	Delivers customer orders, confirms deliveries, and updates the owner regarding order status.	Utilizes the system's delivery tracking module to update delivery progress and confirm completed orders.

Adrian Echepari (Scrum Master)	Oversees project management, sprint planning, and coordination among team members.	Ensures that development aligns with project timelines and user requirements.
Joseph Emmanuel Magno (Backend Developer)	Designs and implements server-side logic, database connections, and API integration.	Develops the backend infrastructure of SariSite, ensuring data accuracy and system reliability.
Cybier Kim Otchia (Frontend Developer)	Designs and codes the user interface for both the customer and admin panels.	Implements responsive and user-friendly web layouts for accessibility and ease of use.
John Cyrus Agoncillo (System Analyst)	Analyzes user needs, defines functional requirements, and models the system workflow.	Bridges the operational needs of Alyn Store and the technical design of the SariSite system.

4.3 System Policies and Procedures

This section outlines the existing manual policies and procedures that guide the daily operations of Alyn Store. These processes reflect how transactions, deliveries, and customer interactions are currently handled without the use of an online system.

4.3.1 Customer Access and Ordering Policy

- Customers may visit the store physically during regular business hours.
- Orders are made verbally at the counter or through handwritten lists provided by customers.
- For home deliveries, orders are submitted through phone calls, text messages, or personal messages.
- No customer authentication or registration is required.
- Orders are processed on a first-come, first-served basis.

4.3.2 Inventory and Data Recording Policy

- Inventory is monitored manually through visual inspection and periodic counting.
- Product quantities and stock levels are remembered by the store owner or written in a notebook.
- All sales, expenses, and restocks are manually recorded in a physical notebook.
- Price changes are updated verbally or written manually by the owner.

4.3.3 Order Handling and Delivery Policy

- Walk-in customers receive purchased items immediately.
- For delivery orders, the store owner receives the details via message or call.
- Deliveries are handled manually by the owner's son.
- Delivery updates are communicated informally (e.g., messages, calls).
- There is no official tracking system or formal documentation of pending/completed deliveries.

4.3.4 Payment and Transaction Policy

- The store accepts cash and GCash as payment methods.
- The customer chooses whether to pay in cash at the store or through GCash before/after delivery.
- The store owner verifies GCash payments by checking the reference number or SMS confirmation.
- All transactions are recorded manually in the notebook, including mode of payment.
- Canceled or incomplete payments are noted verbally or written informally.

4.3.5 Data Security and Privacy Policy

- Customer information such as names, addresses, and contact numbers are received verbally or through messages.
- These details are not stored in a secure system; they remain with the owner and assisting family members.
- All records in notebooks are kept inside the store with no formal security measures.

- There are no established privacy protocols aside from the owner keeping customer details confidential.

4.3.6 Inventory Backup and Maintenance Policy

- No formal backup exists for transaction or inventory records.
- All important information is stored only in a notebook or based on the owner's memory.
- If the notebook is damaged or misplaced, records may be lost permanently.
- Store maintenance—such as restocking or reorganizing shelves—is done manually.

4.3.7 Customer Support and Issue Handling

- Customers report issues (wrong items, missing change, delivery delays) directly to the store owner.
- Complaints are addressed through immediate verbal communication and corrective action.
- There is no structured issue logging or formal documentation process.

4.4 User Operation Procedures

User Role	Procedure	Purpose
Admin / Store Owner	Assist customers → Check stocks manually → Record sales → Collect cash/GCash → Coordinate deliveries	Manage daily store activities and ensure smooth customer transactions
Customer	Visit store or send order → Request items → Pay via cash/GCash → Receive items or wait for delivery	Purchase goods conveniently
Delivery Handler	Receive delivery instructions → Deliver items → Communicate	Provide basic delivery service

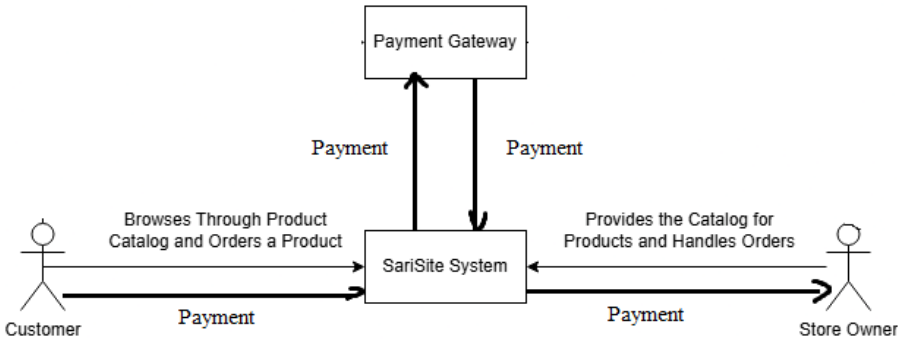
	completion verbally	
Record Keeper (Owner)	Write transactions in notebook → Manually monitor inventory → Review sales for restocking	Maintain store records and stock awareness

4.5 Current Manual Workflow of Alyn Store

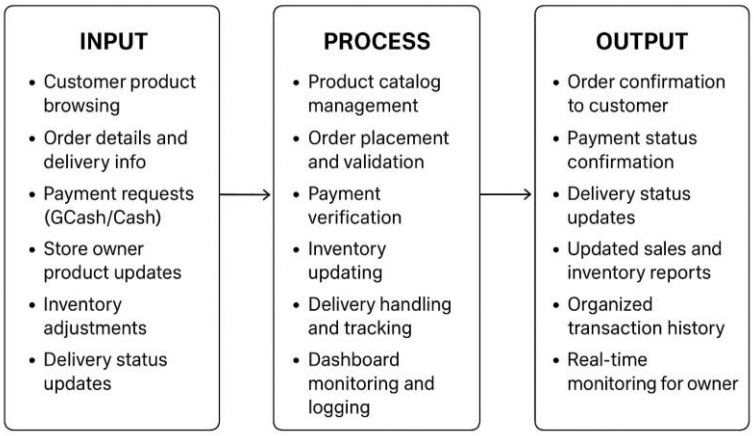
1. Customers arrive at the store or send in orders via phone/message.
2. The owner checks available stock manually.
3. Items are listed manually in a notebook along with the price and total amount.
4. Customers pay using cash or GCash.
5. For deliveries, the owner relays instructions to her son.
6. Deliveries are confirmed informally, with no tracking system.
7. No formal record exists for completed or pending deliveries.
8. Sales summaries and inventory levels rely heavily on the owner's manual notes and observations.
9. Stock replenishment is based on manual counting and visual checking.

4.6 Conceptual Framework

The **SariSite System** aims to transform Alyn Store's manual operations into an automated, web-based ordering and delivery platform. It provides an integrated environment where customers can conveniently browse products, place online orders, and request deliveries. Simultaneously, the store owner can efficiently manage inventory, track orders, and monitor delivery status through the admin dashboard.



The conceptual design of the SariSite System can be interpreted through an Input–Process–Output (IPO) framework, illustrating how data and actions from customers, store owners, and the payment gateway flow through the system to produce meaningful results for store operations and customer service.



4.6.1 Input

The system receives essential inputs from three main stakeholders:

a. Customer Inputs

- Browsing and selecting products
- Placing online orders
- Providing delivery information
- Choosing a payment method (GCash or cash)

b. Store Owner Inputs

- Providing and updating the product catalog
- Managing stock availability
- Accepting or preparing orders
- Updating order and delivery statuses

c. Payment Gateway Inputs

- Processing customer payments
- Sending payment confirmations
- Handling secure transfer of transaction details

These inputs serve as the raw data and actions required for the system to operate properly.

4.6.2 Process

The SariSite System acts as the central platform that processes all interactions:

- Order Management: Receives customer orders and forwards them to the store owner
- Catalog Management: Displays updated product listings for customer browsing
- Inventory Coordination: Ensures that stock information is synchronized with the store owner's input
- Payment Handling: Connects with the payment gateway for payment requests and validation
- Delivery Coordination: Allows the store owner to handle delivery requests and update delivery progress
- System Communication: Facilitates data exchange between customers, store owners, and the payment gateway

This stage transforms the raw inputs into organized, actionable workflows.

4.6.3 Output

The system produces outputs that benefit both customers and the store:

a. Customer Outputs

- Successfully placed orders
- Real-time updates on delivery status
- Confirmed payments
- Access to an organized and easy-to-use product catalog

b. Store Owner Outputs

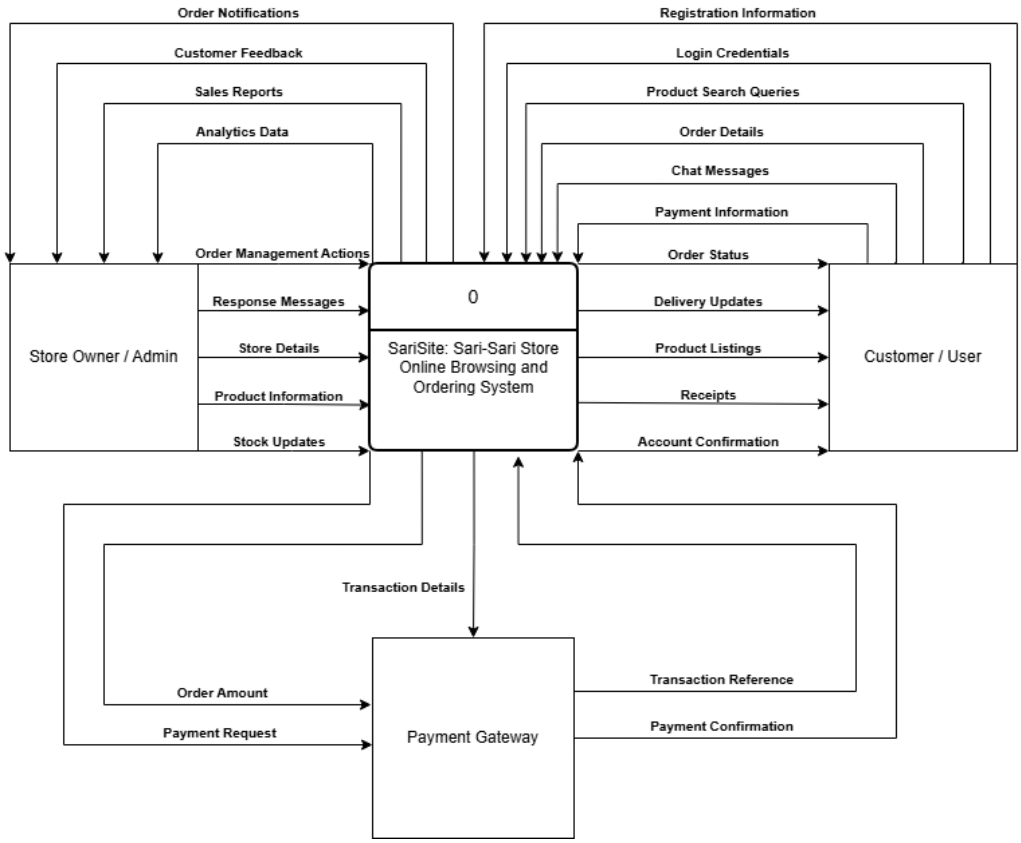
- Centralized dashboard for tracking orders and deliveries
- Verified payment results from the gateway
- Efficient monitoring of sales and inventory

c. System Outputs

- Secure payment confirmations
- Organized transaction records
- Clear communication between all participants

These outputs enhance operational efficiency and improve customer experience.

4.7 Context Data Flow Diagram (Level 0 Description)



Overview

The Context Data Flow Diagram (Level 0) of the SariSite System illustrates the overall interaction between the system and its external entities. The SariSite platform serves as a digital solution that enables customers within a subdivision to browse available products, place online orders, and make payments conveniently. The system connects customers with the store owner, who manages product listings, updates stock availability, and monitors orders through an integrated admin interface. A third-party payment gateway, such as GCash, processes online payments and sends transaction confirmations back to the system.

This diagram presents the SariSite System as a single, unified process that receives input data—such as registration information, product searches, order details, and payments—and generates outputs including order confirmations, receipts, delivery updates, and sales reports. The diagram defines the system's boundaries by showing only

the entities that directly interact with it, excluding any internal components or data stores. This provides a high-level view of how the system operates within its environment and serves both the store owner and customers efficiently.

Chapter V

Software Project Estimation

5.1 Schedule of Activities

The project follows an **Agile Software Development Life Cycle (SDLC)** approach. Development is divided into iterative sprints, each focusing on specific system components such as the frontend, backend, database integration, and testing. This approach allows continuous feedback and improvement throughout the project.

The schedule of activities outlines the estimated duration of each development phase, from planning to deployment. The table below will serve as a reference for monitoring progress and ensuring timely completion of project milestones.

5.1.1 Project Schedule of Activities

Phase	Description	Duration	Remarks
Planning and Requirement Analysis	Identifying user needs, defining scope, and establishing objectives	1 Week	
System Design	Designing the database, architecture, and user interface	1 Month	Conduct meetings and gather user requirements; finalize project scope and objectives.
Development	Coding and implementing system modules	2 Months	Divide tasks among developers; integrate

			front-end and back-end components.
Testing and Debugging	Validating system functionality, performance, and usability	1 Month	Perform unit and integration testing; fix identified errors and improve performance.
Deployment and Evaluation	Deploying the system for actual use and gathering feedback	1 Month	Conduct user acceptance testing (UAT); gather feedback and finalize documentation.

5.1.2 Gantt Chart of SariSite Project Development



5.2 Resource Requirements

The **SariSite System** utilizes cost-effective, open-source, and cloud-based tools to support development and deployment. These resources were chosen to ensure scalability, maintainability, and accessibility during the entire project cycle.

5.3 Development Tools and Environment

The following tools and platforms were used in the development of the system:

- **Programming Languages and Frameworks:** Dart and Flutter (for cross-platform frontend development), Node.js and Express.js (for backend API and server-side operations).
- **Database and Hosting:** MongoDB (for data storage and backend management).
- **Authentication:** Google Cloud Authentication (for secure sign-in and account management).
- **API Testing:** Bruno (for testing and validating backend routes and data flow).
- **Version Control:** GitHub (for collaborative development and version tracking).
- **Development Environment:** Visual Studio Code (for integrated development and debugging).

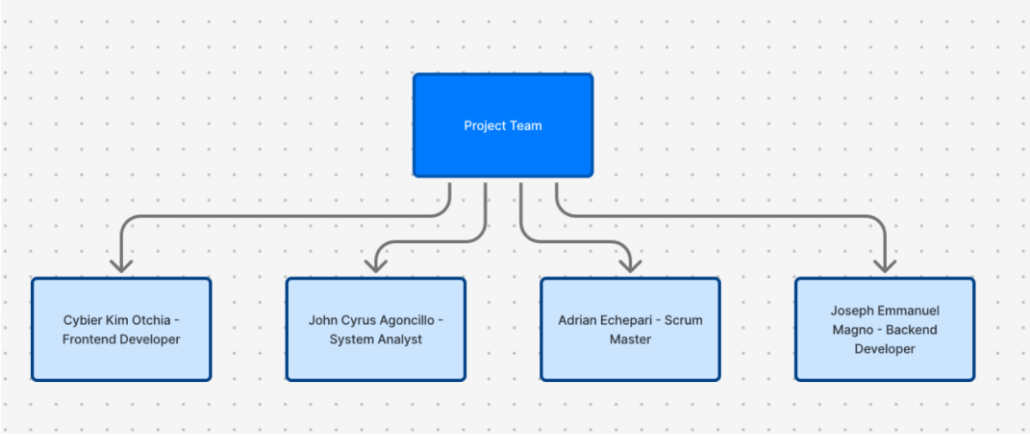
5.4 Human Resources

The system was developed by a team composed of students fulfilling specific roles in system design, development, and management.

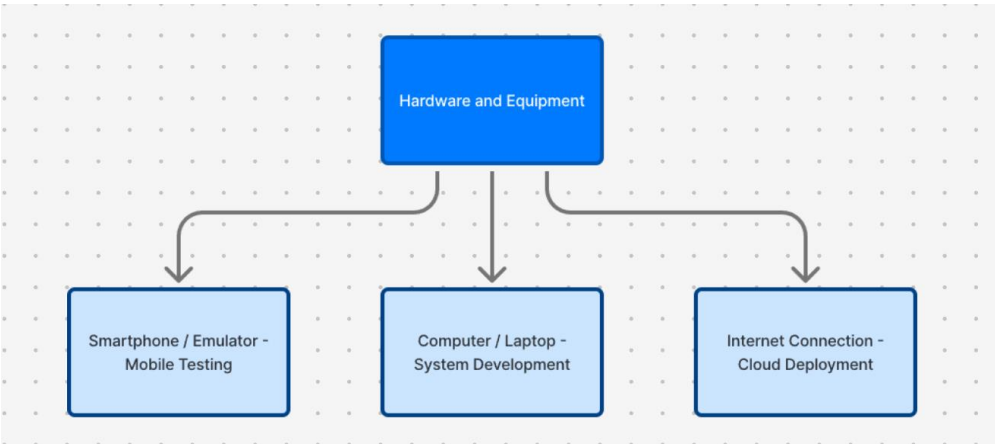
Team Assignments and Workload Distribution

- **Project Manager / Scrum Master**
  Responsible for overseeing project timelines, facilitating team communication, managing sprint activities, and ensuring that development aligned with project goals and store requirements.
- **System Analyst**
  Conducted requirement gathering from the sari-sari store owner, analyzed existing workflows, documented system requirements, and ensured the system design accurately reflected real store operations.
- **Frontend Developer**
  Handled the user interface design and implementation, ensuring that pages such as inventory, product listing, and transaction views were user-friendly, responsive, and properly connected to backend functionalities.
- **Backend Developer**
  Developed core system logic, APIs, and database interactions. Responsible for implementing inventory processes, product CRUD operations, authentication logic, and ensuring data accuracy and security.

5.4.1 Human Resource Allocation



5.5 Hardware and Equipment



5.6 Hardware Requirements

Category	Specification	Purpose
Personal Computer / Laptop	Minimum 8GB RAM, i5 Processor or higher	System development and testing

Stable Internet Connection	At least 10 Mbps	Online collaboration and cloud deployment

5.7 Cost Estimation

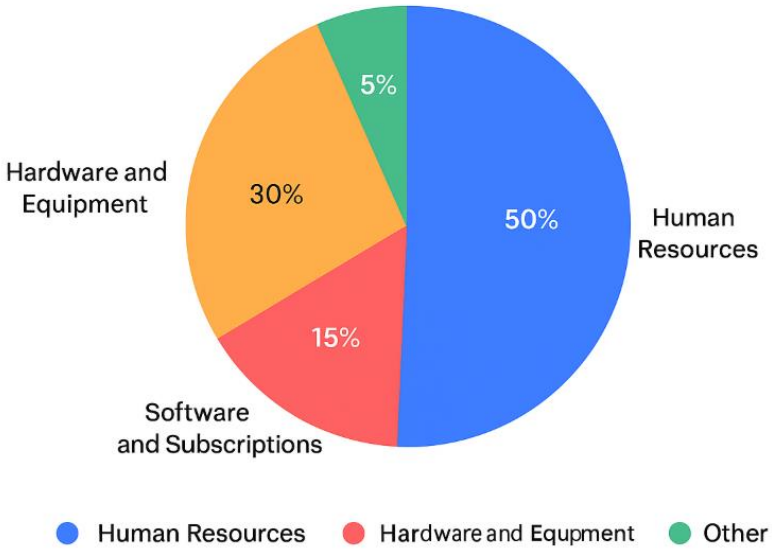
The cost estimation covers all direct and indirect expenses incurred during the development of the SariSite system. Since most of the tools used are free and cloud-based, the projected budget mainly includes operational and documentation costs.

5.7.1 Estimated Project Cost

Category	Details	Estimated Cost (₱)
Direct Costs	Internet load, electricity, cloud hosting, and domain name	**₱3,500**
Indirect Costs	Printing, documentation, research materials, and transportation	**₱1,700**
Contingency	Buffer for unexpected expenses or upgrades	**₱520**
Total Estimated Cost		**₱5,720**

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5.8 Cost Distribution of SariSite Project



Chapter VI

Requirements Analysis and Specification Model

6.1 Process Specification Modelling

6.1.1 Conceptual Design

The proposed **SariSite Online Browsing and Ordering System** aims to modernize the operations of Alyn Store by automating its traditional walk-in purchasing process. The system provides an accessible online platform where customers can browse available products, add items to a shopping cart, and place orders for home delivery. The store owner can monitor, confirm, and manage these orders through a dedicated dashboard, ensuring organized transactions and real-time updates on delivery statuses.

Main Objectives:

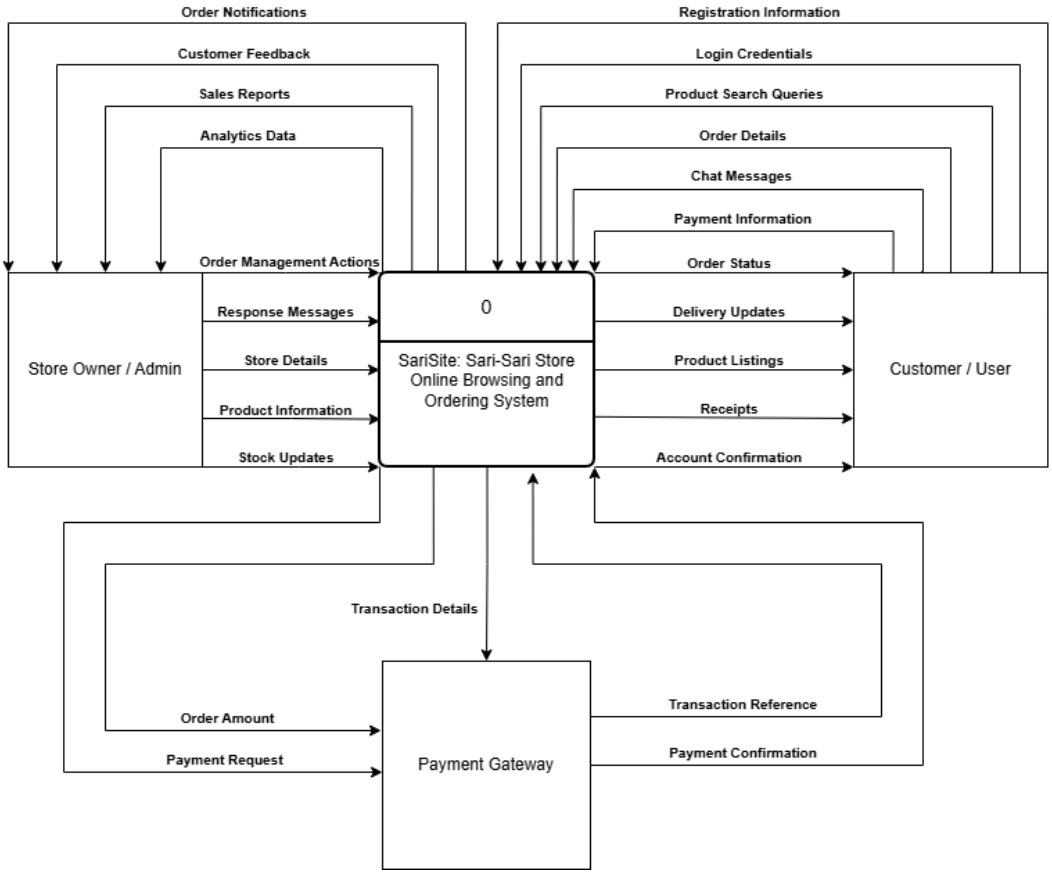
- To automate product ordering and delivery management processes.
- To enhance customer convenience through online access and digital transactions.
- To enable the store owner to efficiently track orders, stock levels, and deliveries.
- To minimize manual record-keeping and reduce errors in order handling.

The system follows a client-server model where the frontend (Flutter) communicates with the backend (Node.js and Express.js) through APIs connected to a MongoDB database. Authentication and security are managed via Firebase Authentication (Google Sign-In), ensuring that only verified users can access their respective accounts.

6.1.2 Data Flow Diagrams

The Data Flow Diagram (DFD) provides a structured visualization of how data moves through the SariSite system—from customers placing online orders to store owner management and delivery confirmation.

Context DFD



Overview:

The diagram illustrates the detailed internal data flow of the **SariSite System**, expanding on the context-level representation. It breaks down the single main process into five interconnected subprocesses that manage user accounts, product listings, orders, payments, and reporting. It also shows how data moves between the **Customer/User**, **Store Owner/Admin**, and the **Payment Gateway**, along with the system's main data stores.

At this level, the system is represented by multiple linked processes (1.0 – 5.0), highlighting both the **inputs** and **outputs** that connect them to internal data stores and external entities.

External Entities

1. Customer / User

- Represents subdivision residents who browse, order, and purchase items from the sari-sari store through the system.
- They can register, log in, browse available products, place orders, make payments, and chat with the store owner.
- They receive real-time responses such as **account confirmations, product listings, delivery updates, order statuses, and receipts**.

Data Flows:

- *Incoming:* Registration Information, Login Credentials, Product Search Queries, Order Details, Chat Messages, Payment Information
- *Outgoing:* Account Confirmation, Product Listings, Delivery Updates, Order Status, Receipts

2. Store Owner / Admin

- Represents the sari-sari store operator responsible for managing products, orders, deliveries, and communication with customers.
- This entity acts as both the **store manager** and the **system administrator**.
- They perform various management tasks and receive performance summaries from the system.

Data Flows:

- *Incoming:* Order, Customer Feedback, Sales Reports, *Outgoing:* Product Information, Stock Updates, Store Details, Response Messages, Order Management Actions

3. Payment Gateway

- Refers to external e-wallet services (e.g., GCash) integrated into the system to process online payments securely.
- It handles payment requests, verifies transaction details, and confirms completed payments.

Data Flows:

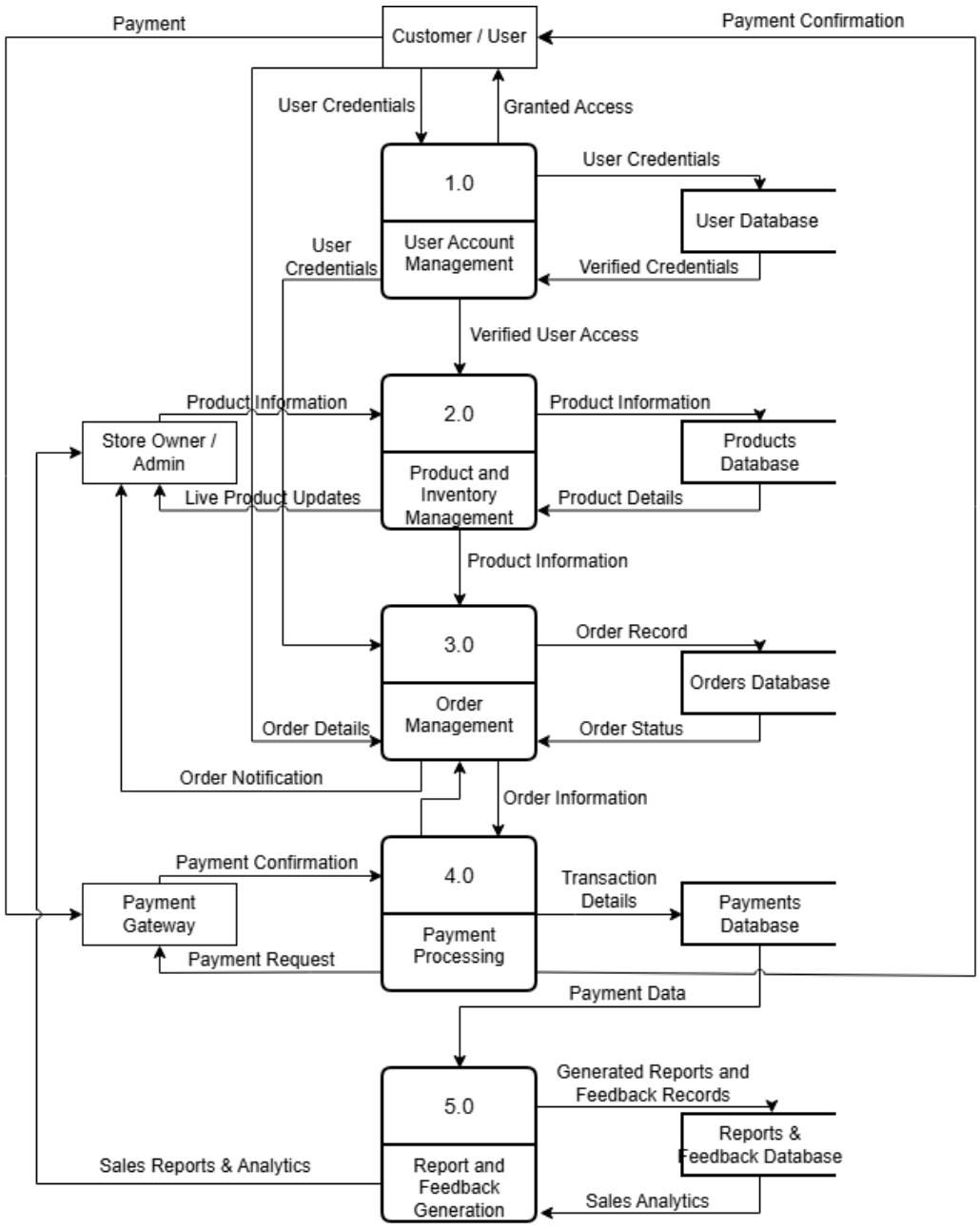
- *Incoming:* Transaction Reference, Payment Confirmation
- *Outgoing:* Order Amount, Payment Request, Transaction Details

Central Process

0 – SariSite: Sari-Sari Store Online Browsing and Ordering System

- Acts as the **core process** that connects customers, store owners, and the payment service.
- It facilitates the entire online transaction cycle—from customer registration, product browsing, and order placement, to payment processing, delivery coordination, and report generation.
- Internally, it manages data validation, order tracking, and chat handling to ensure a seamless user experience.

Top-Down DFD



Explanation:

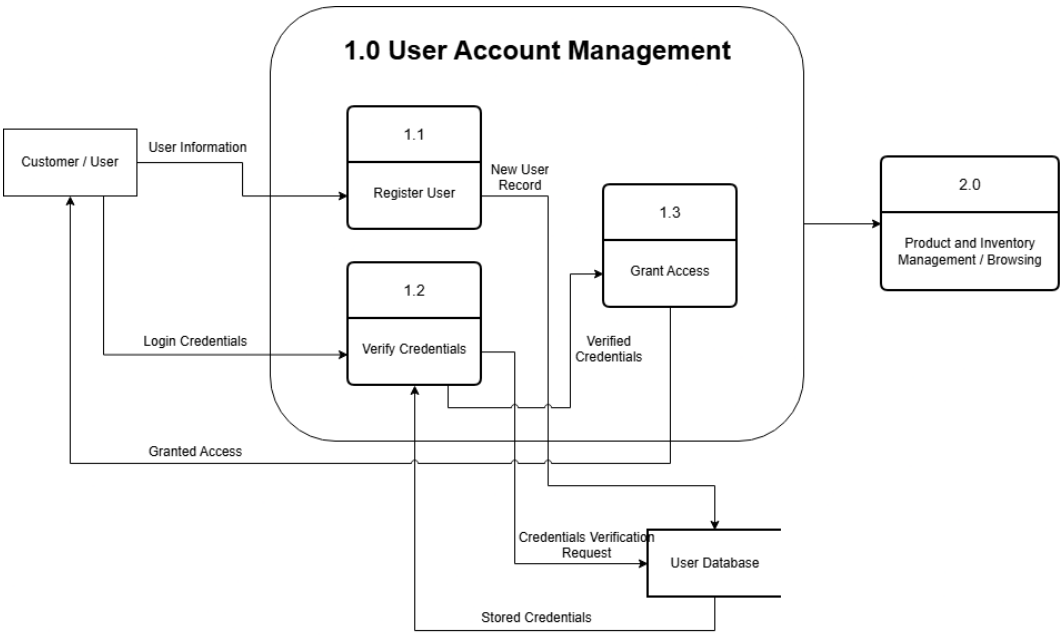
The Top-Down Data Flow Diagram (DFD) of SariSite illustrates the overall data movement and interaction among the system's main processes, external entities, and data stores. The system begins with User Account Management (1.0), where the Customer or User provides their credentials to register or log in. These credentials are verified through the User Database, granting access to the system once authenticated. After gaining access,

the user proceeds to Product and Inventory Management (2.0), where product details are managed by the Store Owner or Admin and retrieved from the Products Database. This process allows live product updates, ensuring that the latest inventory data is displayed to users browsing items.

Once users select items and proceed with checkout, Order Management (3.0) handles the recording and tracking of order details. The system generates an Order Record stored in the Orders Database, allowing both the customer and admin to monitor order status. Order information is then passed to Payment Processing (4.0), where transaction details are managed. The system interacts with the Payment Gateway to process payments and updates the Payments Database accordingly. Payment confirmation or decline messages are returned to both the customer and admin, ensuring transparency in transaction outcomes.

Finally, all transaction and feedback data flow into Report and Feedback Generation (5.0), where the system compiles and analyzes sales, payment, and customer feedback records. These are stored in the Reports and Feedback Database and used performance summaries accessible to the store owner. Through this top-down approach, the DFD provides a comprehensive overview of how SariSite integrates user actions, admin management, data validation, and automated reporting to streamline online ordering and store operations.

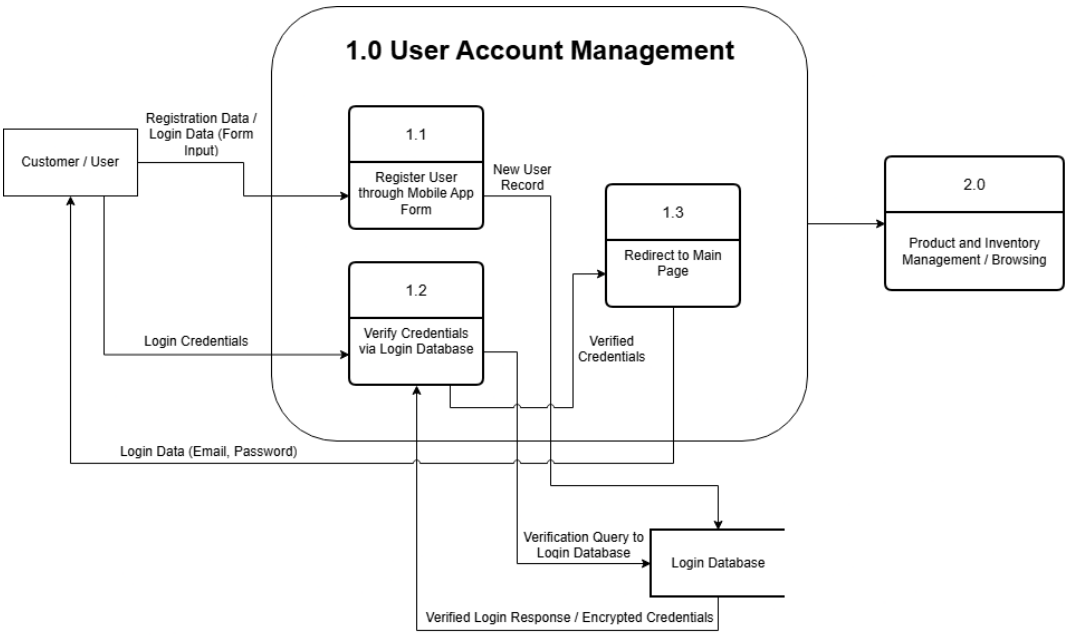
1.0 User Account Management Logical DFD



Explanation:

The Logical Data Flow Diagram (DFD) for User Account Management illustrates how user authentication and access processes are handled within the system. The Customer/User provides their information and credentials, which are sent to Process 1.1 (Register User) for creating a new user record, or to Process 1.2 (Verify Credentials) for authentication. The system checks the provided data against the User Database, retrieving stored credentials to confirm validity. Once verified, Process 1.3 (Grant Access) authorizes the user to access the platform and proceed to other modules such as Product and Inventory Management (2.0). This process ensures that only registered and verified users can enter the system, maintaining secure login and controlled data access throughout the platform.

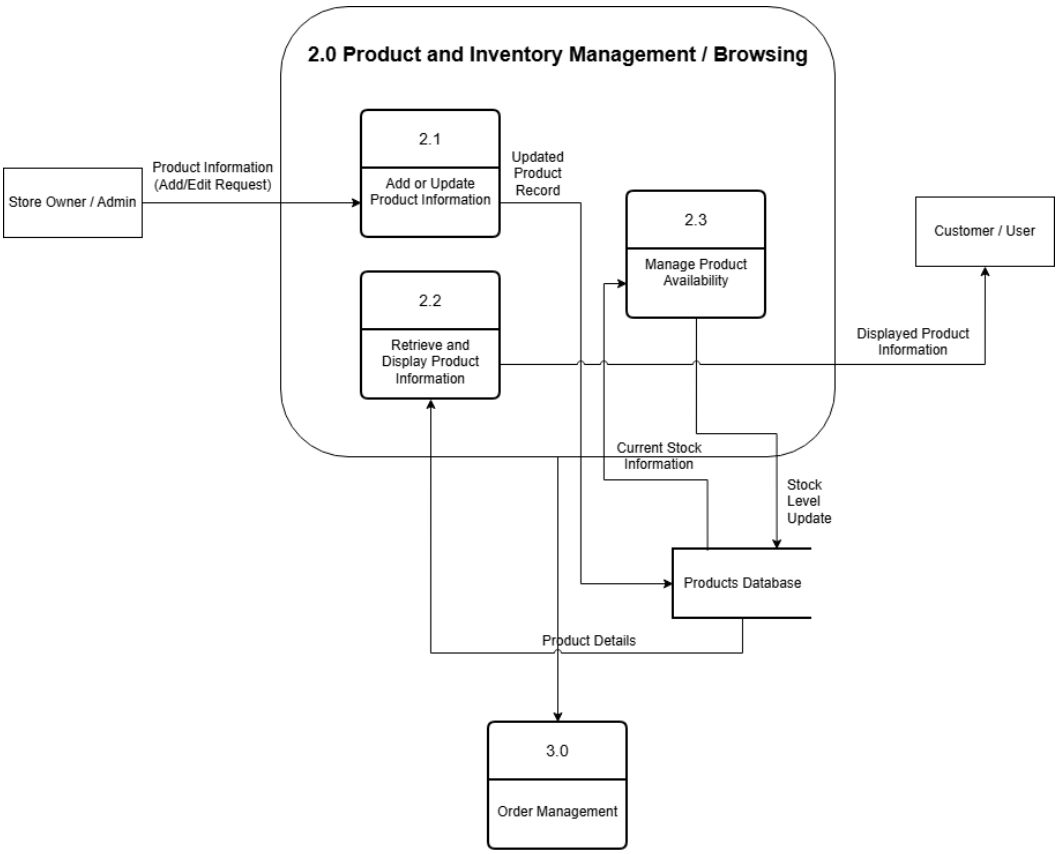
1.0 User Account Management Physical DFD



Explanation:

The Physical Data Flow Diagram (DFD) for User Account Management represents how the actual user authentication and registration processes are implemented in the system. The Customer/User enters their registration details or login credentials through the mobile site form, which are transmitted to Process 1.1 (Register User through Mobile Site Form) for account creation or to Process 1.2 (Verify Credentials via Login Database) for authentication. The system validates this information against the Login Database, where encrypted credentials are stored and verified. Once the user's identity is confirmed, Process 1.3 (Redirect to Main Page) grants access to the application and connects the user to subsequent modules such as Product and Inventory Management (2.0). This physical process emphasizes the use of the mobile interface and real-time database verification to ensure secure, efficient, and user-friendly login operations.

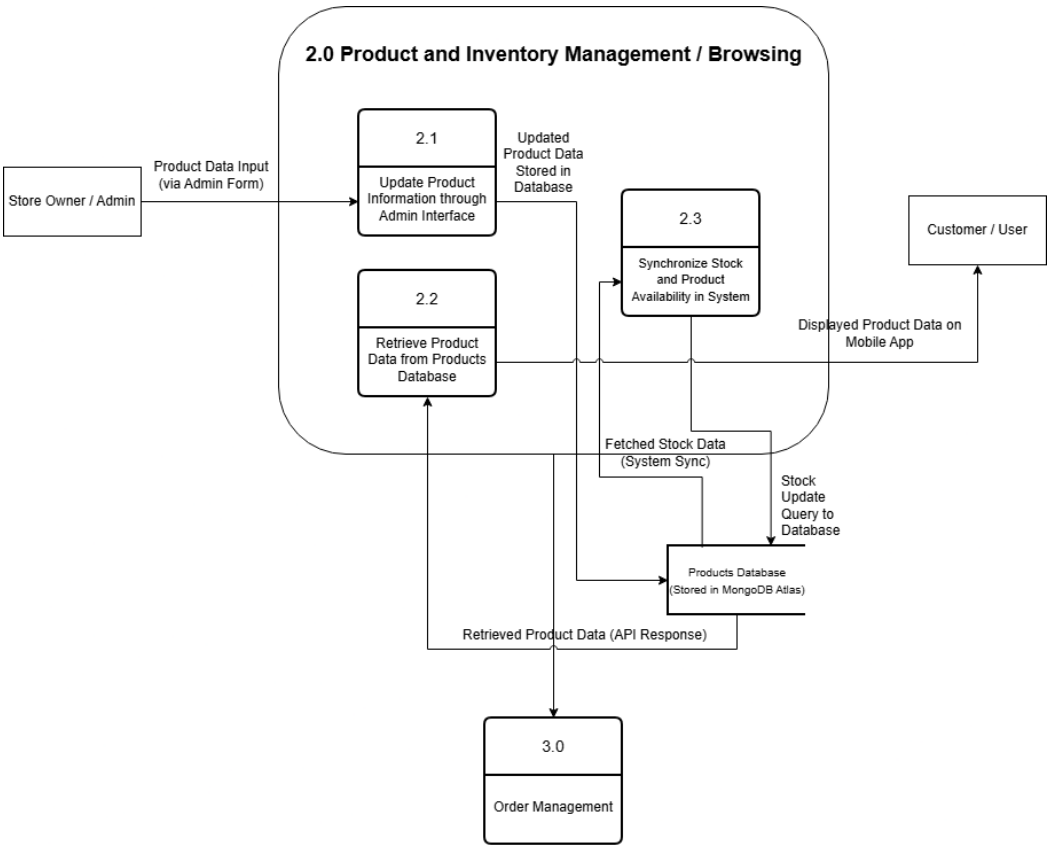
2.0 Product and Inventory Management / Browsing Logical DFD



Explanation:

The Logical Data Flow Diagram (DFD) for Product and Inventory Management / Browsing illustrates how product information is maintained, updated, and displayed within the system. The Store Owner or Admin submits product data or edit requests to Process 2.1 (Add or Update Product Information), which stores updated product records in the Products Database. The stored data is then accessed by Process 2.2 (Retrieve and Display Product Information) to present accurate product listings and details to the Customer/User. Meanwhile, Process 2.3 (Manage Product Availability) monitors current stock levels and updates availability status in the database to ensure that out-of-stock items are properly flagged. This logical process ensures that both users and administrators can interact with consistent, up-to-date product data while maintaining accurate inventory control across the system.

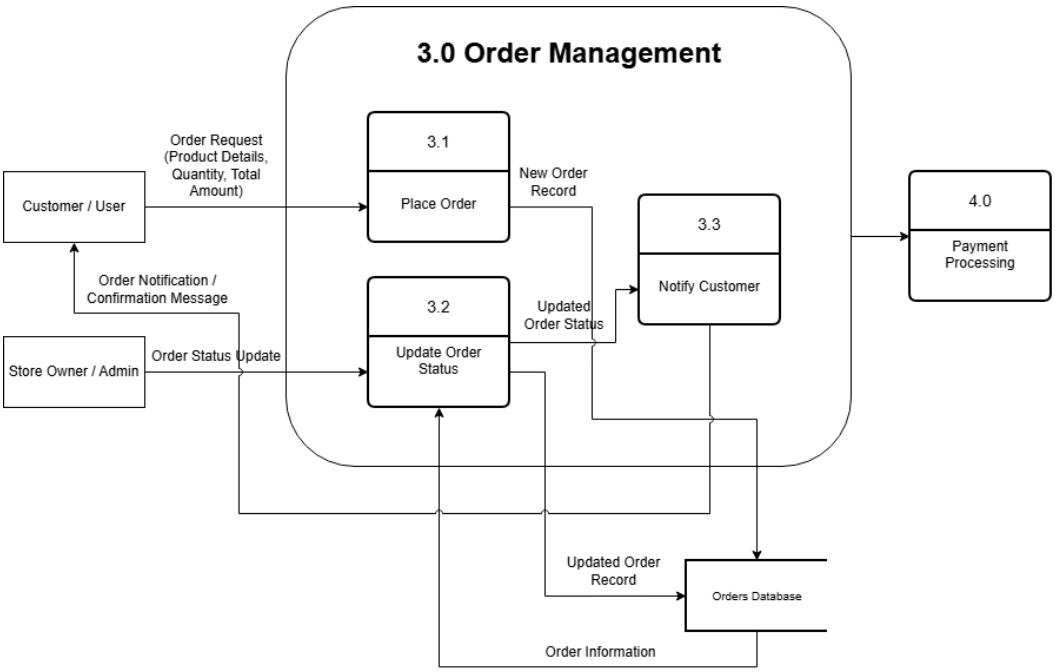
2.0 Product and Inventory Management / Browsing Physical DFD



Explanation:

The Physical Data Flow Diagram (DFD) for Product and Inventory Management / Browsing demonstrates how the system implements real-time product updates and display functions through connected interfaces and databases. The Store Owner or Admin inputs or modifies product details via the Admin Interface, which are processed by Process 2.1 (Update Product Information through Admin Interface) and stored in the Products Database (stored in MongoDB Atlas). Process 2.2 (Retrieve Product Data from Products Database) fetches the updated data and sends it to the site for user display, ensuring that customers view the most recent product listings. Meanwhile, Process 2.3 (Synchronize Stock and Product Availability in System) automatically tracks and updates stock quantities through a system sync mechanism, keeping inventory data consistent between the database and the application. This physical model emphasizes the system's integration of live database synchronization and user-facing interfaces to maintain accurate, up-to-date product information.

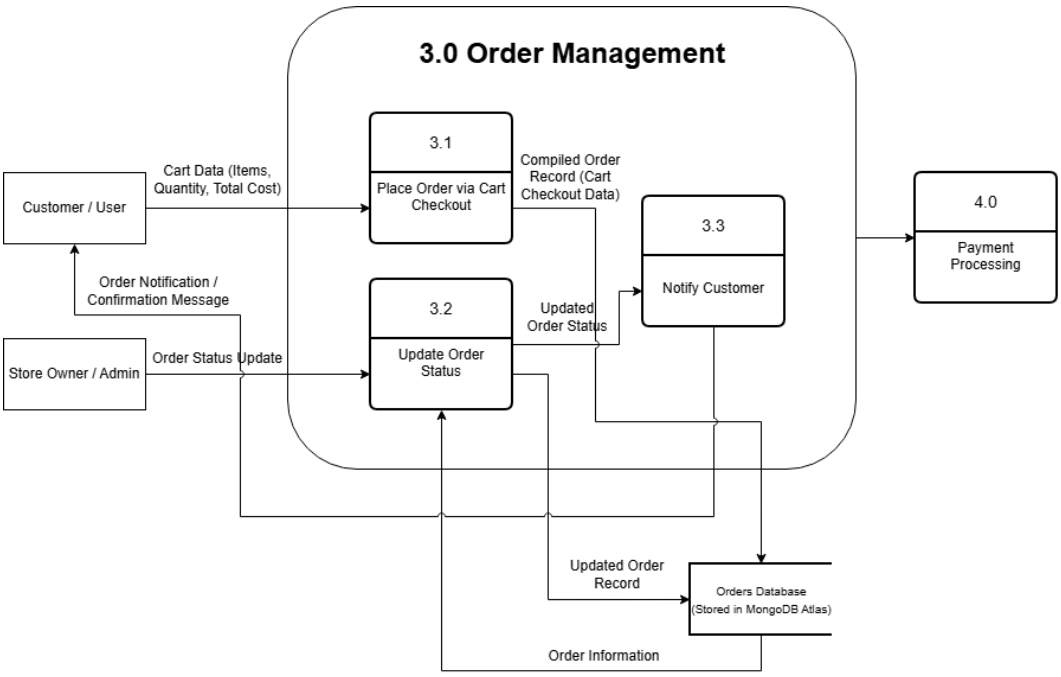
3.0 Order Management Logical DFD



Explanation:

The Logical Data Flow Diagram (DFD) for Order Management outlines the process of handling customer orders within the system. The Customer/User initiates an order by submitting product details, quantity, and total amount to Process 3.1 (Place Order), which generates a New Order Record stored in the Orders Database. Process 3.2 (Update Order Status) ensures that the status of each order is regularly monitored and modified based on transaction progress or store owner updates. Meanwhile, Process 3.3 (Notify Customer) sends order confirmations and status updates to both the customer and the Store Owner/Admin. This logical flow ensures accurate order tracking, real-time updates, and efficient communication between users and administrators throughout the ordering process.

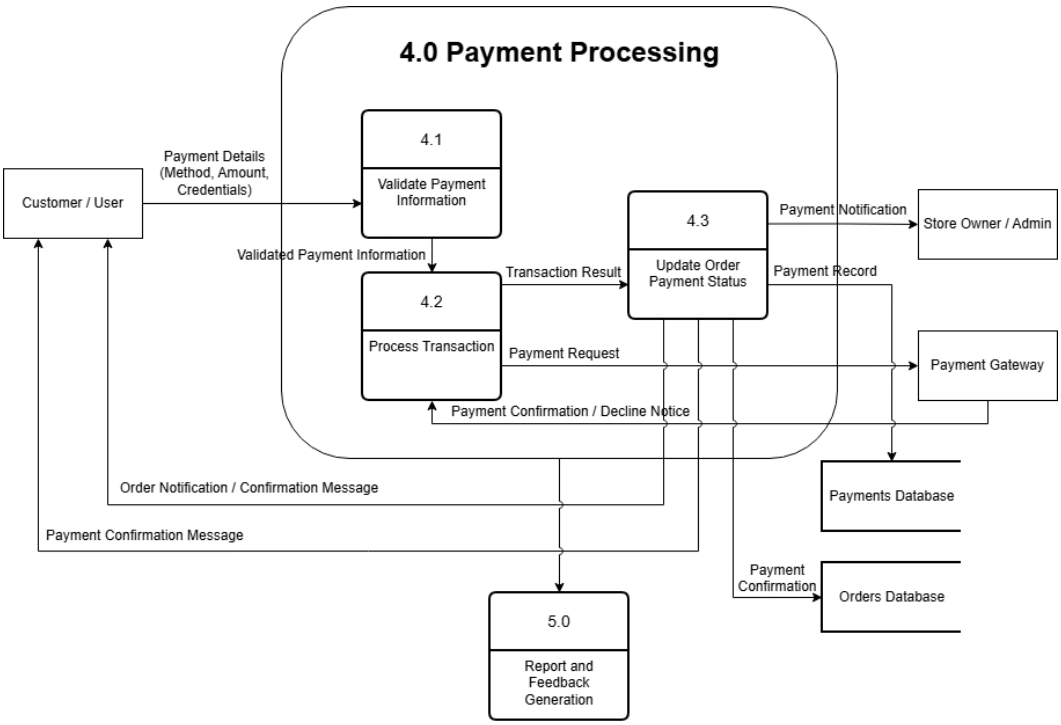
3.0 Order Management Physical DFD



Explanation:

The Physical Data Flow Diagram (DFD) for Order Management illustrates how the system processes and manages customer orders using the actual interfaces and database components. The Customer/User selects products and proceeds to checkout, sending Cart Data (items, quantity, total cost) to Process 3.1 (Place Order via Cart Checkout), where all selected items are compiled into a single order record. The compiled data is stored in the Orders Database (stored in MongoDB Atlas) for tracking and processing. Process 3.2 (Update Order Status) allows the Store Owner/Admin to modify the order's progress, while Process 3.3 (Notify Customer) sends confirmation and order status updates back to the customer. This physical flow highlights the interaction between the mobile checkout system, database synchronization, and admin-side updates to ensure that each order is accurately recorded, monitored, and communicated in real time.

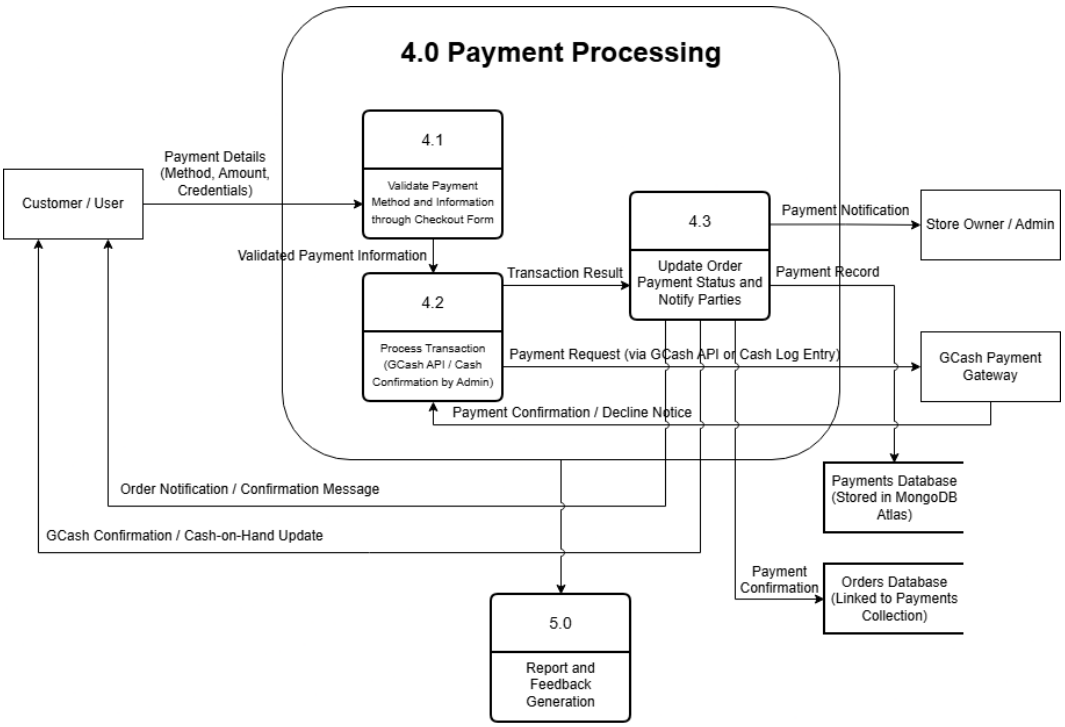
4.0 Payment Processing Logical DFD



Explanation:

The Logical Data Flow Diagram (DFD) for Payment Processing shows how the system manages payment verification, transaction handling, and order confirmation. The Customer/User provides Payment Details (method, amount, and credentials), which are sent to Process 4.1 (Validate Payment Information) for verification before proceeding to Process 4.2 (Process Transaction). The system then communicates with the Payment Gateway to process the transaction and records the payment results in the Payments Database. Once the transaction is completed, Process 4.3 (Update Order Payment Status) updates both the Orders Database and notifies the Store Owner/Admin of the payment confirmation or decline. The system also sends a Payment Confirmation Message to the customer and forwards verified transaction data to Report and Feedback Generation (5.0). This logical model ensures that all payment activities are accurately validated, recorded, and communicated across users and system components.

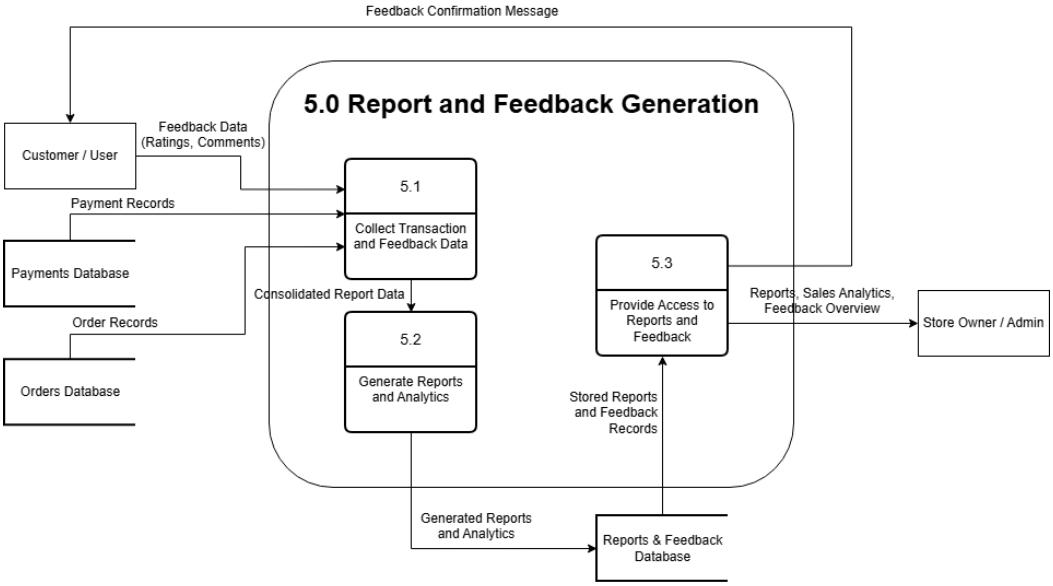
4.0 Payment Processing Physical DFD



Explanation:

The Physical Data Flow Diagram (DFD) for Payment Processing illustrates how the system handles payments using actual interfaces and technologies such as GCash and Cash-on-Hand confirmation. The Customer/User submits Payment Details (method, amount, and credentials) through the checkout form, which are validated in Process 4.1 (Validate Payment Method and Information through Checkout Form). Once validated, Process 4.2 (Process Transaction via GCash API or Cash Confirmation by Admin) executes the transaction and sends the result to Process 4.3 (Update Order Payment Status and Notify Parties). Payment data and results are stored in the Payments Database (stored in MongoDB Atlas) and linked to the Orders Database for record synchronization. The Store Owner/Admin and the customer both receive notifications of payment confirmation or decline, while GCash Payment Gateway handles real-time transactions. This physical flow demonstrates how automated digital payments and manual cash validations are integrated into the system to ensure secure and accurate financial processing.

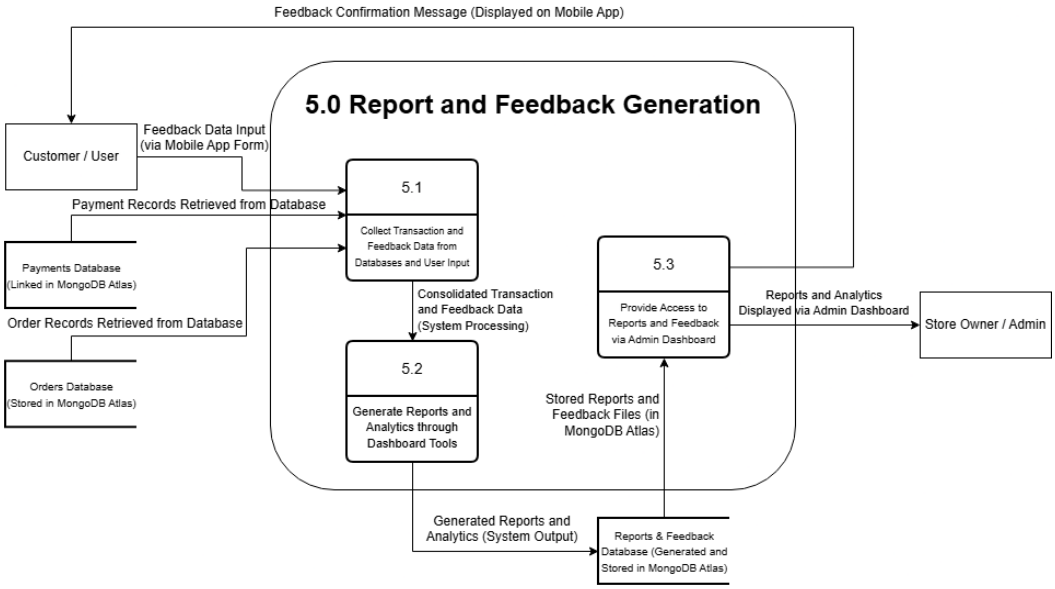
5.0 Report and Feedback Generation Logical DFD



Explanation:

The Logical Data Flow Diagram (DFD) for Report and Feedback Generation illustrates how the system collects and processes transaction and feedback data to produce reports for store management. The Customer/User submits Feedback Data (ratings and comments), while Payment Records and Order Records are retrieved from the Payments Database and Orders Database, respectively. These inputs are consolidated in Process 5.1 (Collect Transaction and Feedback Data) and analyzed in Process 5.2 (Generate Reports) to produce detailed summaries of sales performance and customer satisfaction. The generated data is stored in the Reports & Feedback Database and accessed through Process 5.3 (Provide Access to Reports and Feedback), which delivers reports to the Store Owner/Admin. A Feedback Confirmation Message is also sent to the user, ensuring that their input is acknowledged. This logical model ensures accurate data collection, analysis, and reporting to support better decision-making and service improvement.

5.0 Report and Feedback Generation Physical DFD



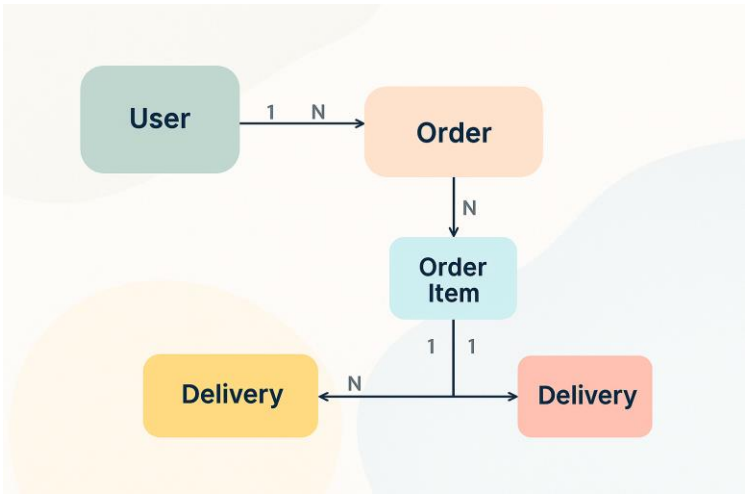
Explanation:

The Physical Data Flow Diagram (DFD) for Report and Feedback Generation illustrates how the system uses actual interfaces and databases to collect, analyze, and present data for reporting purposes. The Customer/User submits Feedback Data Input (via the mobile site form), while Payment Records and Order Records are retrieved from the Payments Database and Orders Database, both stored in MongoDB Atlas. These inputs are consolidated in Process 5.1 (Collect Transaction and Feedback Data from Databases and User Input) and processed through Process 5.2 (Generate Reports through Dashboard Tools) to create summaries of store performance, sales, and user satisfaction. The generated reports are stored in the Reports & Feedback Database (generated and stored in MongoDB Atlas) and accessed through Process 5.3 (Provide Access to Reports and Feedback via Admin Dashboard), which displays the results to the Store Owner/Admin. A Feedback Confirmation Message is also displayed on the customer's mobile site to acknowledge successful submission. This physical data flow demonstrates how the system integrates database queries, dashboard tools, and mobile interfaces to automate reporting and enhance feedback management.

6.2 Data Specification Modelling

6.2.1 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) defines the logical structure of the SariSite database, including its key entities, attributes, and relationships.



Entities and Attributes:

- User (user_id, name, email, role, password)
- Product (product_id, name, price, quantity, category)
- Order (order_id, user_id, total_amount, order_date, status)
- OrderItem (order_item_id, order_id, product_id, quantity, subtotal)
- Delivery (delivery_id, order_id, address, delivery_status, delivery_date)

Relationships:

- A **User** can place multiple **Orders**.
- An **Order** can contain multiple **OrderItems**.
- Each **OrderItem** is linked to a specific **Product**.
- Each **Order** is associated with one **Delivery** record.

6.2.2 Data Dictionary

1. **Users Collection**

Field Name	Data Type	Size	Description
user_id	ObjectID	-	Unique identifier for each user
name	String	100	User's full name
email	String	100	User's email address
role	String	50	User role (e.g., Customer, Store Owner)
password	String	255	Encrypted password (if using email login; ignored for Google Sign-In)

2. Products Collection

Field Name	Data Type	Size	Description
product_id	ObjectId	-	Unique identifier for each product
name	String	100	Product name
price	Double	-	Product price
quantity	Int	-	Available stock count
category	String	50	Product category/classification
description	String	255	Short details about the product
shelf_life	String	50	Expiry date or shelf-life (e.g., "6 months")
net_weight	Decimal	10,2	The weight of the product itself (excluding packaging) in grams
variation	String	100	Product variations (size, brand, pack type, etc.)

3. Orders Collection

Field Name	Data Type	Size	Description

order_id	ObjectId	-	Unique order identifier
user_id	ObjectId	-	References the ordering user
total_amount	Double	-	Total computed cost of the order
order_date	Date	-	Date and time when the order was placed
status	String	50	Order status (Pending, Confirmed, Preparing, Delivered, etc.)

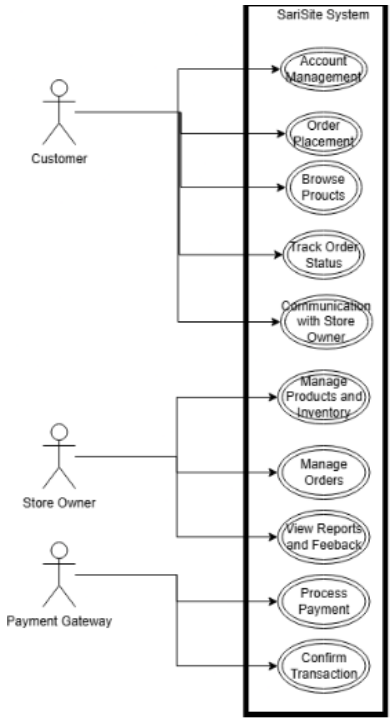
4. Order Items Collection

Field Name	Data Type	Size	Description
order_item_id	ObjectId	-	Unique identifier for each order item
order_id	ObjectId	-	References the parent order
product_id	ObjectId	-	References the product purchased
quantity	Int	-	Quantity of the product ordered
subtotal	Double	-	Computed amount for this specific item

5. Delivery Collection

Field Name	Data Type	Size	Description
delivery_id	ObjectId	-	Unique identifier for each delivery record
order_id	ObjectId	-	References the order being delivered
address	String	255	Customer's delivery address
delivery_status	String	50	Current delivery status (Pending, Out for Delivery, Delivered)
delivery_date	Date	-	Actual delivery completion date

6.2.3 UML (Unified Modeling Language)



The Use-Case Diagram illustrates the core interactions within the SariSite System, showing how the different actors—Customer, Store Owner, and Payment Gateway—engage with the system's major functionalities. The Customer interacts with the system through several key use cases, including Account Management, Order Placement, Browse Products, Track Order Status, and Communication with Store Owner, representing the complete shopping and order-monitoring experience. Meanwhile, the Store Owner accesses administrative and operational features such as Manage Products and Inventory, Manage Orders, and View Reports and Feedback, which support store operations and decision-making. Lastly, the Payment Gateway connects to the system to Process Payment and Confirm Transaction, ensuring secure and verified financial transactions. Together, these interactions show a comprehensive workflow that supports user account handling, product browsing, order processing, communication, inventory management, reporting, and payment verification within the SariSite online ordering and delivery platform.

Chapter VII

Software Design Specification

7.1 Process Specification

This section presents the detailed logic, rules, and algorithms used by the main processes of the SariSite system. Each process is described using structured English following the required guideline format.

P1.0 – User Login and Authentication

Process ID: P1.0

Description: Verifies user credentials using Firebase Authentication and grants access to the system.

Input Data Flows:

- email
- password / Google Sign-In token

Output Data Flows:

- login_status
- user_profile

Business Rules:

1. Only registered users can log in.
2. All passwords must be encrypted.
3. Account must exist in the database before access is granted.

Logic (Structured English):

```
IF user selects Google Sign-In THEN
    SEND token to Firebase Authentication
    IF token is valid THEN
        RETRIEVE user_profile from database
        DISPLAY dashboard page
    ELSE
        DISPLAY "Login Failed"
ENDIF

IF user enters email and password THEN
    VALIDATE email and password format
    CHECK credentials in database
    IF match FOUND THEN
        GRANT user access
    ELSE
        DISPLAY "Invalid Credentials"
    ENDIF
ENDIF
```

P2.0 – Manage Product and Inventory

Process ID: P2.0

Description: Handles adding, updating, and retrieving product information.

Input Data Flows:

- product_name
- price
- quantity
- category

Output Data Flows:

- updated_product_list
- stock_status

Business Rules:

1. Only the Admin/Store Owner can modify products.
2. Quantity must be a non-negative number.
3. Product updates must be saved immediately to MongoDB.

Logic:

```
RECEIVE product update request from admin
CHECK all fields for completeness
IF fields are valid THEN
    UPDATE product record in database
    RETURN updated_product_list
ELSE
    DISPLAY error message
ENDIF
```

P3.0 – Order Placement and Processing

Process ID: P3.0

Description: Processes customer orders and records order details.

Input Data Flows:

- selected_items
- quantity
- total_amount
- user_id

Output Data Flows:

- order_confirmation
- order_status

Business Rules:

1. Order cannot be placed if stock is insufficient.
2. Order ID must be generated uniquely.
3. Orders must be recorded before sending confirmation.

Logic:

```
RECEIVE selected_items and quantity
FOR each item
    CHECK stock availability
    IF insufficient stock THEN
        DISPLAY "Out of Stock"
        CANCEL order
    ENDIF
ENDFOR
CREATE new order_record in database
SEND order_confirmation to user
SET order_status = "Pending"
```

P4.0 – Payment Processing (GCash or Cash)

Process ID: P4.0

Description: Validates and records payment transactions.

Input Data Flows:

- payment_method
- payment_reference (for GCash)
- total_amount

Output Data Flows:

- payment_status
- updated_order_status

Business Rules:

1. GCash payment must be validated through API.
2. Cash is marked “To Pay on Delivery or Pick-up.”

3. Orders cannot be delivered without payment confirmation (except Cash).

Logic:

```
IF payment_method = "GCash" THEN
    SEND payment_reference to GCash API
    IF valid THEN
        SET payment_status = "Paid"
        UPDATE order_status
    ELSE
        DISPLAY "Payment Failed"
    ENDIF
ENDIF

IF payment_method = "COD" THEN
    SET payment_status = "Pay on Delivery"
    UPDATE order_status = "Pending Payment"
ENDIF
```

P5.0 – Delivery Management

Process ID: P5.0

Description: Updates and tracks the delivery status of orders.

Input Data Flows:

- order_id
- delivery_status

Output Data Flows:

- delivery_updates
- delivery_confirmation

Business Rules:

1. Delivery personnel can only update assigned orders.
2. Delivery confirmation requires customer acknowledgment.

Logic:

```
RECEIVE delivery update request
VALIDATE order_id
UPDATE delivery_status in database
NOTIFY customer with updated delivery status
IF status = "Delivered" THEN
    REQUEST customer confirmation
ENDIF
```

7.2 Data Specification

The following tables define the structure and requirements for each collection in the system. Each table lists the field name, data type, whether the field is required, additional rules or notes for proper usage, and an example value. This standardized specification ensures data consistency, supports validation, and serves as a guide for developers when implementing the database and connecting it with application logic.

1. **Users Table**

Field	Type	Required?	Rules/Notes	Example
user_id	STRING	Yes	Unique identifier for each user (MongoDB ObjectId)	64f9c1a2b3d4e5f678901234
name	STRING	Yes	User's full name	John Doe
email	STRING	Yes	User's email address	johndoe@example.com
role	STRING	Yes	User role (e.g., Customer, Store Owner)	Customer
password	STRING	Yes	Encrypted password (if using email login; ignored for Google Sign-In)	bcrypted...

2. **Products Table**

Field	Type	Required?	Rules/Notes	Example
product_id	STRING	Yes	Unique identifier for each product (MongoDB ObjectId)	64f9c1a2b3d4e5f678901235
name	STRING	Yes	Product name	Chippy
price	DOUBLE	Yes	Product price	20.00
quantity	INT	Yes	Available stock count	30
category	STRING	Yes	Product category/classification	Snacks
description	STRING	No	Short details about the product	Crispy Corn Chips
shelf_life	STRING	No	Expiry date or shelf-life (e.g., “6 months”)	12 months
net_weight	DECIMAL	No	Weight of the product itself (excluding packaging) in grams	11 grams
variation	STRING	No	Product variations (size, brand, pack type, etc.)	BBQ Flavor

3. Orders Table

Field	Type	Required?	Rules/Notes	Example
order_id	STRING	Yes	Unique order identifier (MongoDB ObjectId)	64f9c1a2b3d4e5f678901236
user_id	STRING	Yes	References the ordering user	64f9c1a2b3d4e5f678901234

total_amount	DOUBLE	Yes	Total computed cost of the order	1599.99
order_date	DATE	Yes	Date and time when the order was placed	2025-11-24T14:30:00
status	STRING	Yes	Order status (Pending, Confirmed, Preparing, Delivered, etc.)	Pending

4. Order Items Table

Field	Type	Required?	Rules/Notes	Example
order_item_id	STRING	Yes	Unique identifier for each order item	64f9c1a2b3d4e5f678901237
order_id	STRING	Yes	References the parent order	64f9c1a2b3d4e5f678901236
product_id	STRING	Yes	References the product purchased	64f9c1a2b3d4e5f678901235
quantity	INT	Yes	Quantity of the product ordered	2
subtotal	DOUBLE	Yes	Computed amount for this specific item	224.00

5. Delivery Table

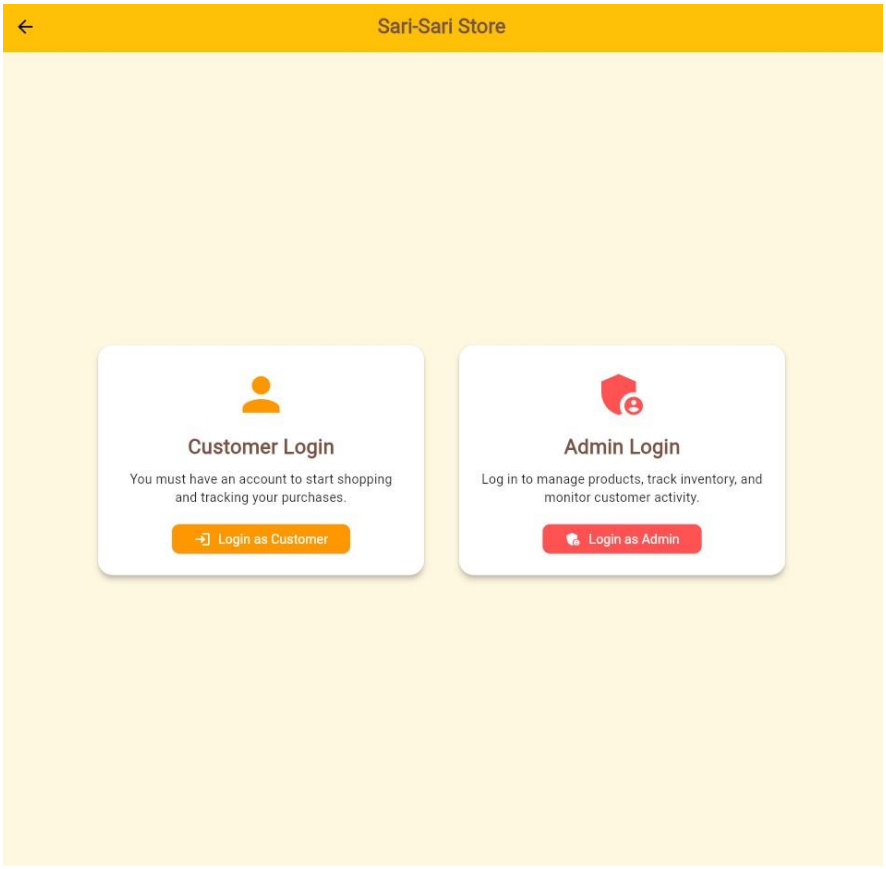
Field	Type	Required?	Rules/Notes	Example
delivery_id	STRING	Yes	Unique identifier for each delivery record	64f9c1a2b3d4e5f678901238

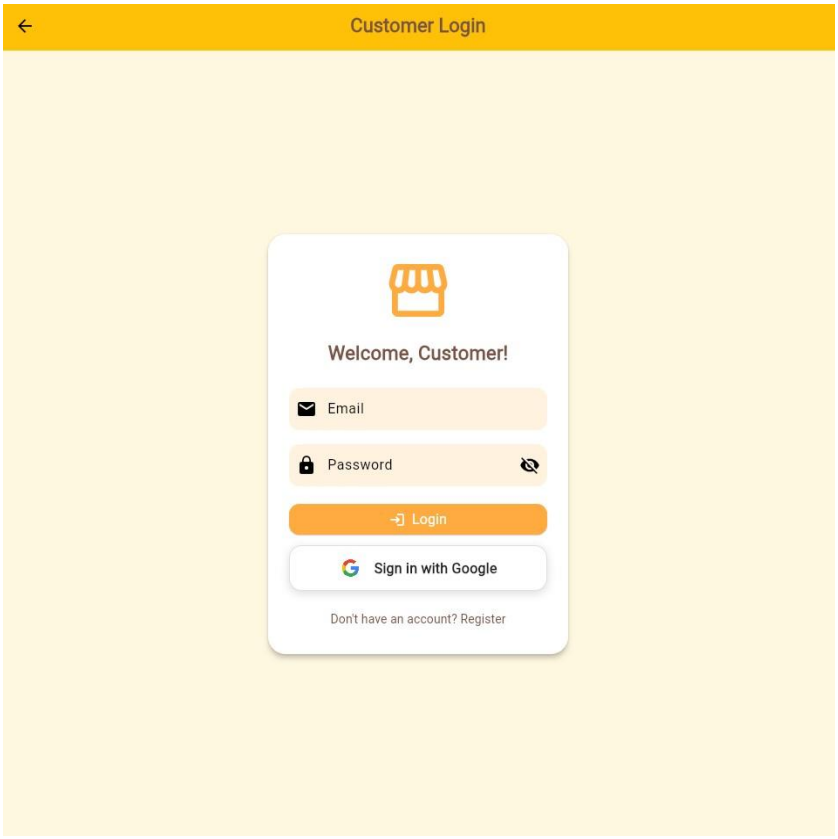
order_id	STRING	Yes	References the order being delivered	64f9c1a2b3d4e5f678901236
address	STRING	Yes	Customer's delivery address	123 Main St, Manila
delivery_status	STRING	Yes	Current delivery status (Pending, Out for Delivery, Delivered)	Out for Delivery
delivery_date	DATE	No	Actual delivery completion date	2025-11-25T10:00:00

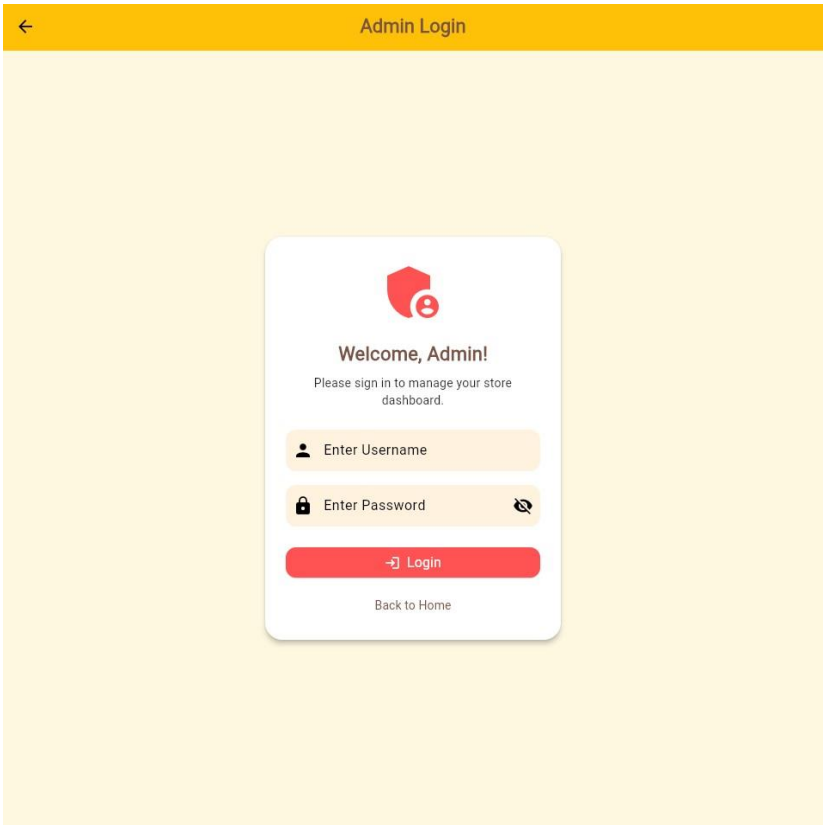
7.3 Screen / Interface Specification

This section describes the appearance and behavior of major system screens.

7.3.1 User Login Screen




Customer Login
Welcome, Customer!
Email
Password
Login
Sign in with Google
Don't have an account? Register


Admin Login
Welcome, Admin!
Please sign in to manage your store dashboard.
Enter Username
Enter Password
Login
Back to Home

Screen ID: SCR-LOGIN-01

Purpose: Allows users to log in using email/password or Google Sign-In.

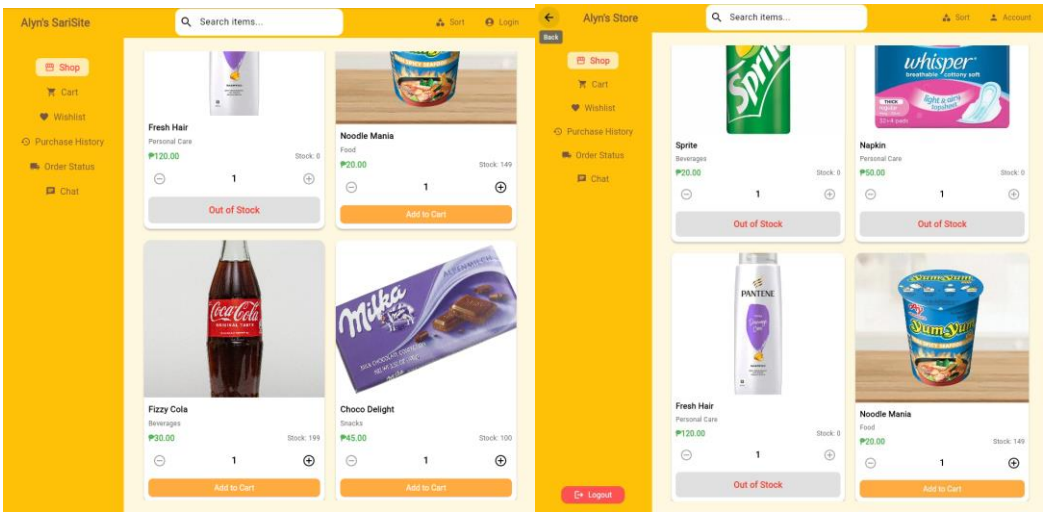
Description:

- Fields: Email, Password
- Buttons: Login, Google Sign-In
- Error prompts displayed under fields.

User Actions:

- Enter credentials
- Click Login
- Redirects to Dashboard

7.3.2 Customer Dashboard



Logged Out Logged In

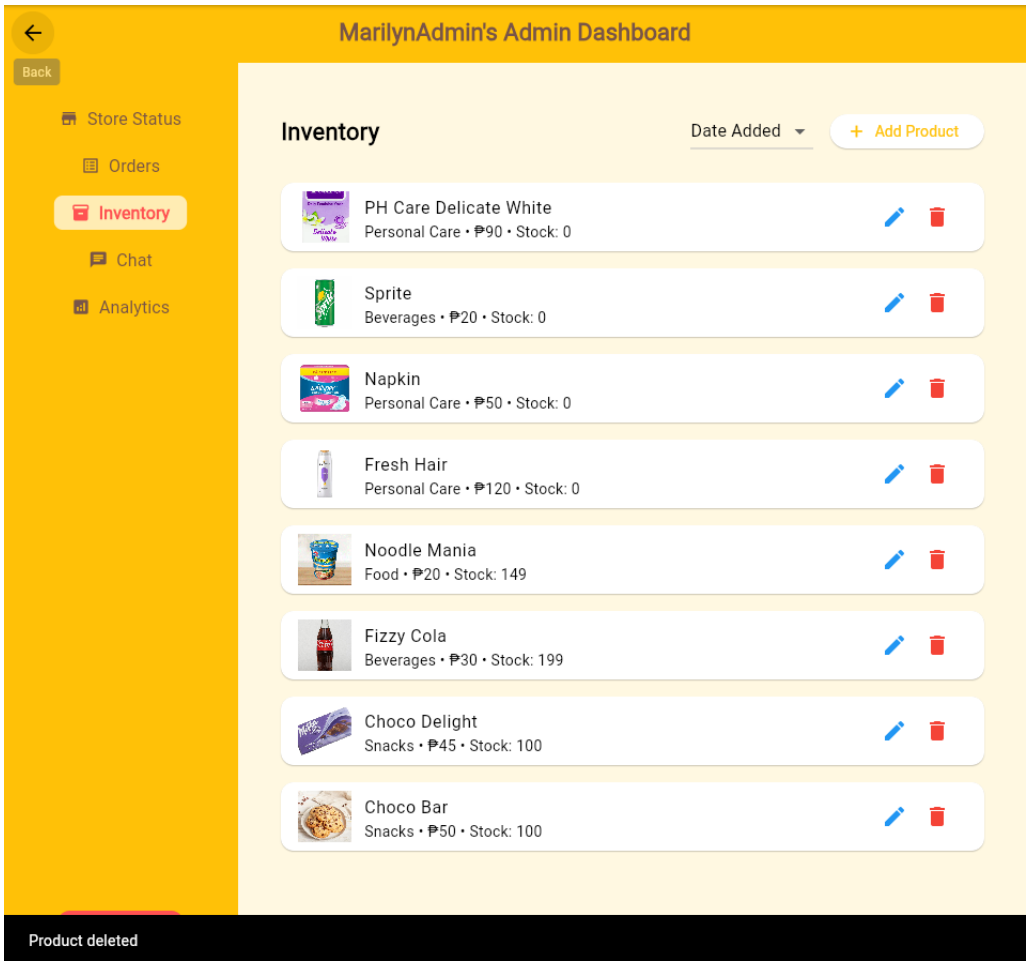
Screen ID: SCR-CUST-02

Purpose: Provides access to product browsing and order functions.

Components:

- Navigation bar (Products, My Orders, Profile)
- Product grid
- Search bar

7.3.3 Admin Dashboard



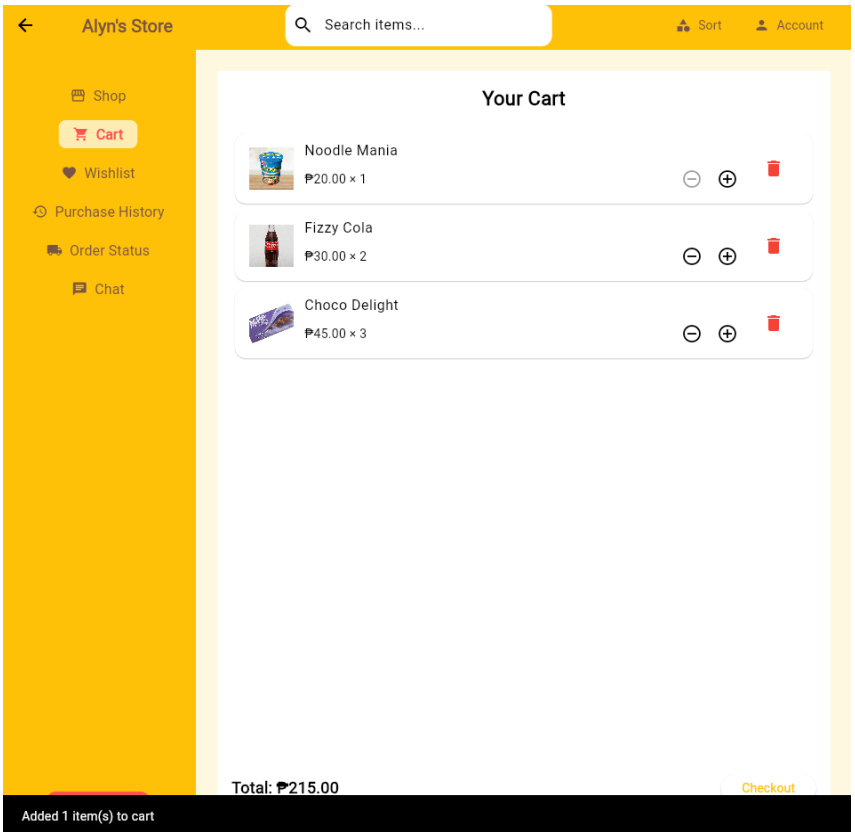
Screen ID: SCR-ADMIN-03

Purpose: Allows the owner to manage products, orders, and delivery.

Modules:

- Product Management
- Order List
- Sales Report

7.3.4 Cart and Checkout Screen



Screen ID: SCR-PAY-04

Purpose: Allows customers to select payment method and confirm order.

Fields:

- Item Summary
- Total Amount
- GCash / COD options

7.4 Program Module Specification

7.4.1 General Information

Project Identification:

- **Project Name:** SariSite: Enhancing Accessibility of Localized Sari-Sari Stores in Philippine Subdivision Through Online Ordering

- **Sponsor / Client:** Marilyn Otchia (Owner of the Alyn's Sari-sari Store)
- **Project Manager:** Adrian V. Echepari (Scrum Master)

Business Problem Description:

Sari-sari stores in the area rely on manual record-keeping methods such as notebooks or spreadsheets to track sales, inventory, and expenses. This process is error-prone, time-consuming, and lacks real-time visibility into stock levels and sales performance. Consequently, store owners face challenges in maintaining accurate inventory, identifying best-selling items, and generating reports for decision-making.

Project Purpose:

The SariSite system is designed to centralize sales and inventory management for sari-sari stores in a simple, web-based application. It aims to streamline product management, track sales, manage orders, and provide actionable insights to store owners, reducing manual work while improving operational efficiency.

Project Scope:

- **In-Scope:**
    - Web-based system for inventory, sales, and order tracking.
    - User roles: Admin (store owner) and Customer (optional for online orders if implemented).
    - Inventory management with automated stock updates.
    - Basic product management: add, edit, and remove items.
- **Out-of-Scope:**
    - Integration with external e-commerce platforms.
    - Complex AI or recommendation systems.
    - Logistics or delivery management beyond the store's physical location.

7.4.2 Project Requirements

SariSite is a modular, web-based application that enables sari-sari store owners to manage inventory, sales, and orders efficiently. The system uses a client-server architecture

with a React.js or Flutter web front-end, Node.js back-end, and MongoDB for persistent storage. It provides secure user authentication, real-time inventory tracking, and reporting tools to support informed business decisions.

Product Features:

- **User Authentication & Management:**
  - Secure registration and login for store owners.
  - Role-based access: Admin can manage products and view reports.
- **Product & Inventory Management:**
  - CRUD operations for products: add, edit, delete, view.
  - Stock levels automatically updated after each sale.
  - Alerts for low stock items to prevent stockouts.
- **Sales & Order Tracking:**
  - Capture all sales transactions with timestamps and product details.
  - Generate invoices or receipts for each transaction.
  - Support for multiple payment types: cash, digital payments (optional).
- **Reporting & Dashboard:**
  - Real-time sales and inventory dashboards.
  - Reports for best-selling products, stock levels, and revenue trends.
  - Export options for Excel or PDF reports.
- **System Administration:**
  - Centralized control over users, products, and sales data.
  - Role-based restrictions to prevent unauthorized access.

System Features:

- **User Authentication:**
  - Secure login with hashed passwords.
  - Session management and role-based access control.
- **Product Management:**
  - CRUD operations with product details (name, price, stock, category, description).
  - Validation rules to prevent negative stock levels.
- **Sales & Transaction Handling:**
  - Automatic stock adjustment upon recording a sale.

- Track transaction history for auditing purposes.

- **Dashboard & Reporting:**
  - Visual charts for sales, revenue, and inventory.
  - Filtering and search functionality for products and transactions.

Nonfunctional Requirements:

- **Performance:**
  - Handle multiple concurrent users without significant slowdown.
  - Fast response times for product lookup and transaction processing.
- **Reliability & Availability:**
  - Ensure system uptime during store operating hours.
  - Backup and recovery mechanisms for data protection.
- **Security:**
  - Encrypt sensitive information such as passwords.
  - Prevent unauthorized access via role-based controls and secure API endpoints.
- **Scalability:**
  - Support growth in the number of products, transactions, or stores.
  - Modular backend to accommodate new features with minimal disruption.
- **Maintainability:**
  - Modular code structure to simplify future updates.
  - Centralized logging for error tracking and debugging.
- **Usability:**
  - Intuitive user interface for store owners.
  - Consistent feedback and validation messages to guide user actions.
- **Data Integrity:**
  - Ensure accurate inventory updates for every transaction.
  - Maintain historical transaction data for auditing and reporting purposes.
- **Compliance & Constraints:**
  - All sensitive data stored securely using best practices.
  - System must be deployable on common web servers with minimal configuration.

Chapter VIII

Software Implementation

8.1 Programming Considerations, Issues, and Tools

8.1.1 Programming Considerations

A. Languages and Frameworks

- Flutter Web is used for the frontend to provide a responsive, cross-platform web application.
- Node.js with Express.js serves as the backend API layer.
- MongoDB Atlas stores all application data, including users, products, orders, and inventory records.
- Firebase Authentication handles secure user registration and login, providing role-based access control.
- MongoDB database is used for storing product images using the Multer package from Node.js.

B. Security Considerations

- All communication between client and server uses HTTPS.
- Session tokens are short-lived and validated on each request.
- Only authorized users are allowed to access the database.
- Sensitive information, including API keys and database credentials, is stored in environment variables.

C. File Handling and Uploads

- Product images are validated for type and file size on both client and server.
- Images are uploaded to cloud storage, with only the URL stored in the database.
- Retry logic ensures reliable file upload in case of network issues.

D. Performance and Testing

- API response times for browsing products, managing inventory, and processing orders are optimized to be under 2 seconds.
- Testing includes:

- Unit tests for individual modules (authentication, products, cart, orders).
- Integration tests to verify communication between frontend, backend, and database.
- User Acceptance Testing (UAT) to confirm end-to-end workflows.

E. Operational Considerations

- Checkout and inventory updates are idempotent to prevent duplicate records or negative stock.
- Logs are maintained for all CRUD operations for auditing and traceability.

F. Monitoring and Maintenance

- Backend errors are centrally logged, with alerts for critical failures.
- Logs support auditing for inventory changes, orders, and user activities.

8.1.2 Programming Issues

- Connectivity Issues: Network problems may disrupt file uploads or web access.
- Token Expiration: Expired session tokens may cause login failures.
- Data Integrity: Simultaneous updates to inventory must be handled to prevent negative stock.
- File Size & Storage: Large product images could exceed storage limits or slow uploads.

8.1.3 Tools and Technologies

Tool / Technology	**Purpose**
Flutter Web / React.js	Frontend web application development
Node.js + Express	Backend API and server management
MongoDB Atlas	Data storage for users, products, orders
Firebase Authentication	User login, registration, and role-based access

Cloud Storage (Firebase / Cloudinary)	Storing uploaded product images
HTTPS	Secure client-server communication
Logging & Alerts	Backend monitoring and operational alerts
Unit / Integration / UAT Testing	System testing and quality assurance

8.2 System Requirements Specification

8.2.1 Hardware Requirements

A. Development Machines (per developer)

- `Minimum: Dual-core CPU, 8 GB RAM, 256 GB SSD/HDD, Windows 10/11, stable internet (≈10 Mbps)`
- Recommended: Quad-core CPU, 16 GB RAM, 512 GB SSD, Windows 10/11, USB debugging cable

B. Testing Devices

- At least one modern web browser (Chrome, Firefox, Edge)
- Optional physical Android/iOS devices if using Flutter Web for mobile view testing

C. Hosting / Server Needs (MVP)

- For MVP, MongoDB Atlas and Firebase Hosting are sufficient
- Ensure enough cloud storage quota for product images (~5 GB for initial deployment)

D. Network & Peripherals

- Stable internet connection for development, testing, and deployment

8.2.2 Software Requirements

A. Core Platform & Frameworks (MVP)

- Frontend: Flutter Web / React.js
- Backend: Node.js + Express
- Database: MongoDB Atlas
- Authentication & Storage: Firebase Authentication and Cloud Storage
- APIs / Third-party Services: None required for MVP

B. Developer Tools & IDEs

- Visual Studio Code or Android Studio
- Git & GitHub for version control

C. Non-functional Software Requirements

- Performance: Page load and database operations under 3 seconds
- Scalability: Designed for small-scale usage (hundreds of products, tens of users)
- Security: Enforce Firebase rules; store credentials securely in environment variables
- Maintainability: Modular code structure, clear README, setup instructions

D. Required Accounts & Keys

- Firebase account for authentication and hosting
- MongoDB Atlas account
- GitHub account

8.2.3 Human Resource Requirements

Role	Description	Required Skills	Responsibilities
Project Manager / Scrum Master	Oversees development process, ensures agile methodology is followed	Agile project management, leadership, stakeholder communication	Manage backlog, facilitate sprint planning, remove impediments
Frontend Developer	Implements web interface and user experience	Flutter / React.js, web UI/UX	Develop pages, integrate APIs, ensure responsive design
Backend Developer	Manages server, database, and business logic	Node.js, Express, MongoDB	Create RESTful API endpoints, manage inventory and order processing

System Analyst	Ensures requirements are met	Analytical and testing skills	Document requirements, conduct functional and integration testing

8.3 Software Testing

8.3.1 Testing Activities

- Unit Testing: Verify individual modules (auth, products, cart, orders) work correctly.
- Integration Testing: Ensure smooth communication between frontend, backend, and database.
- System Testing: End-to-end testing of core workflows (add product → sell → update inventory).
- User Acceptance Testing (UAT): Validate system usability and business requirements with store owner.

8.3.2 Sample Test Cases

Test Case ID	Module	Scenario	Test Steps	Expected Result	Status
TC-AUTH-001	Authentication	Login with valid credentials	Enter email/password → Log in	User redirected to homepage	PASS
TC-PROD-001	Product Management	Add new product	Fill form → Save	Product appears in catalog	PASS
TC-CART-001	Shopping Cart	Add item to cart	Select product → Add to cart	Item added to cart	PASS
TC-ORD-001	Order	Complete	Place order →	Inventory	PASS

	Processing	checkout	Confirm payment	updated, order recorded	

8.4 Installation Process

8.4.1 System Prerequisites

- Backend: Node.js, MongoDB Atlas account, environment variables configured
- Frontend: Web browser (Chrome / Edge / Firefox)
- Development Tools: VS Code or Android Studio, Git

8.4.2 Backend Deployment

1. Push backend code to GitHub
2. Deploy Node.js app to Render / Heroku
3. Configure environment variables: MONGODB_URI, PORT
4. Verify API endpoint returns JSON

8.4.3 Frontend Deployment

1. Build web app (flutter build web or React npm build)
2. Host on Firebase Hosting or Netlify
3. Configure API URL to point to deployed backend

8.4.4 Post-Installation Verification

1. Verify homepage loads products
2. Test login and registration
3. Test adding/editing products
4. Test order workflow and inventory updates
5. Verify error handling for offline mode or invalid inputs

Rollback Procedure

- Revert to previous GitHub commit if deployment fails
- Restore backup of MongoDB database if needed

Chapter IX

Software Maintenance Plan

9.1 Release and Deployment Strategy

9.1.1 Release Cadence

To maintain a balance between rapid improvements and system stability for the SariSite MVP, the following release cadence will be used:

A. Weekly Hotfixes (as needed)

Weekly hotfixes will be carried out to address urgent issues that may arise during the operation of SariSite, such as critical bugs, system errors, login problems, or inventory miscalculations. These hotfixes are first thoroughly tested in the development or staging environment to ensure they effectively resolve the issue without introducing new problems. Once verified, the fixes are promptly deployed to the production environment to prevent any disruptions to business operations or customer activity. This approach ensures that critical issues are addressed quickly, minimizing downtime and maintaining a smooth user experience.

B. Bi-Weekly Sprint Releases (Minor Updates)

SariSite will follow a bi-weekly sprint release schedule for minor updates and improvements. Each sprint lasts two weeks, culminating in a release that combines completed non-breaking features, user interface and experience updates, and planned bug fixes. These releases undergo careful testing in the staging environment to confirm functionality and stability, and deployment to production requires approval from the store owner or project supervisor. This regular schedule allows the development team to consistently enhance the system while maintaining reliability and minimizing operational risk.

C. Final Major Release (Project Handover)

At the conclusion of the project, a final major release will be delivered as the fully tested MVP of SariSite for handover to the client. This release will have undergone extensive testing, including both staging and limited user acceptance testing, to ensure all features function correctly and meet the documented requirements. Deployment notes, system documentation, and user guides will accompany the final release to facilitate an organized and seamless transition to production use. This handover ensures that the client receives a stable and operational system ready for daily business operations.

Conclusion

The release strategy for SariSite combines rapid response and structured improvements to maintain system stability and support business operations. Weekly hotfixes provide immediate resolution for critical issues, bi-weekly sprint releases ensure continuous enhancements and feature delivery, and the final major release offers a thoroughly tested, documented version for handover. This approach balances agility and reliability, with clear procedures in place for deployment and rollback, ensuring the system remains functional, secure, and maintainable throughout its lifecycle.

9.2 Environment Strategy

9.2.1 Development Environment (Local)

- **Purpose:** Code development, unit testing, and initial feature validation by team members.
- **Infrastructure:**
    1. Team workstations: personal laptops or school PCs
    2. Local Flutter Web or React.js development server (flutter run or npm start)
    3. Local Node.js API server (localhost:5000)
    4. MongoDB Atlas test cluster for data
- **Deployment Process:**
    1. Developer writes code on local machine.
    2. Test functionality locally (hot reload for frontend changes).
    3. Commit working code to GitHub feature branch.

4. Create pull request for team review.

- **Access:** Development team only.

9.2.2 Production Environment

- **Purpose:** Live deployment for demonstration, client use, and evaluation.
- **Infrastructure:**
    1. Backend API: Render.com / Heroku web service (Node.js)
    2. Frontend: Hosted on Firebase Hosting or Netlify
    3. Database: MongoDB Atlas production cluster
    4. Authentication: Firebase Authentication (production project)
- **Deployment Process:**
    1. Merge approved pull request to main branch.
    2. Render/Netlify auto-deploys backend/frontend from main branch.
    3. Verify deployment: login → product catalog → order placement → inventory update.
- **Access:** Development team, client/store owner, selected testers.

9.3 Version Control Strategy (GitHub)

Branching Model:

- main branch: Production-ready code (protected, requires PR approval)
- dev branch: Integration branch for ongoing development
- Feature branches: feature/product-management, feature/order-flow, etc.

Promotion Flow:

Local Development → Feature Branch → Dev Branch → Main Branch → Production

Rollback Procedures:

- **Backend:** Revert specific commits in GitHub and redeploy to Render.
- **Database:** Restore MongoDB Atlas snapshots to recover users, products, and orders.
- **Frontend:** If needed, redeploy previous stable build to Firebase Hosting or Netlify.
- **Validation Testing Checklist:**

- Verify login, registration, and authentication workflows
- Check product listing, editing, and deletion
- Test inventory updates after orders
- Confirm order processing and checkout functions
- Ensure error handling displays proper messages for invalid inputs or missing data

9.4 Rollback Plan

Scope:

- Provides a systematic strategy for SariSite to recover from critical production issues while ensuring data integrity for inventory, orders, and user management.

Timeline:

- Planned response within 30 minutes to several hours, depending on issue severity.

Triggers:

- Login or authentication failures
- Broken or malfunctioning product management features
- Corrupted product, user, or order data
- Failed order processing or inventory updates

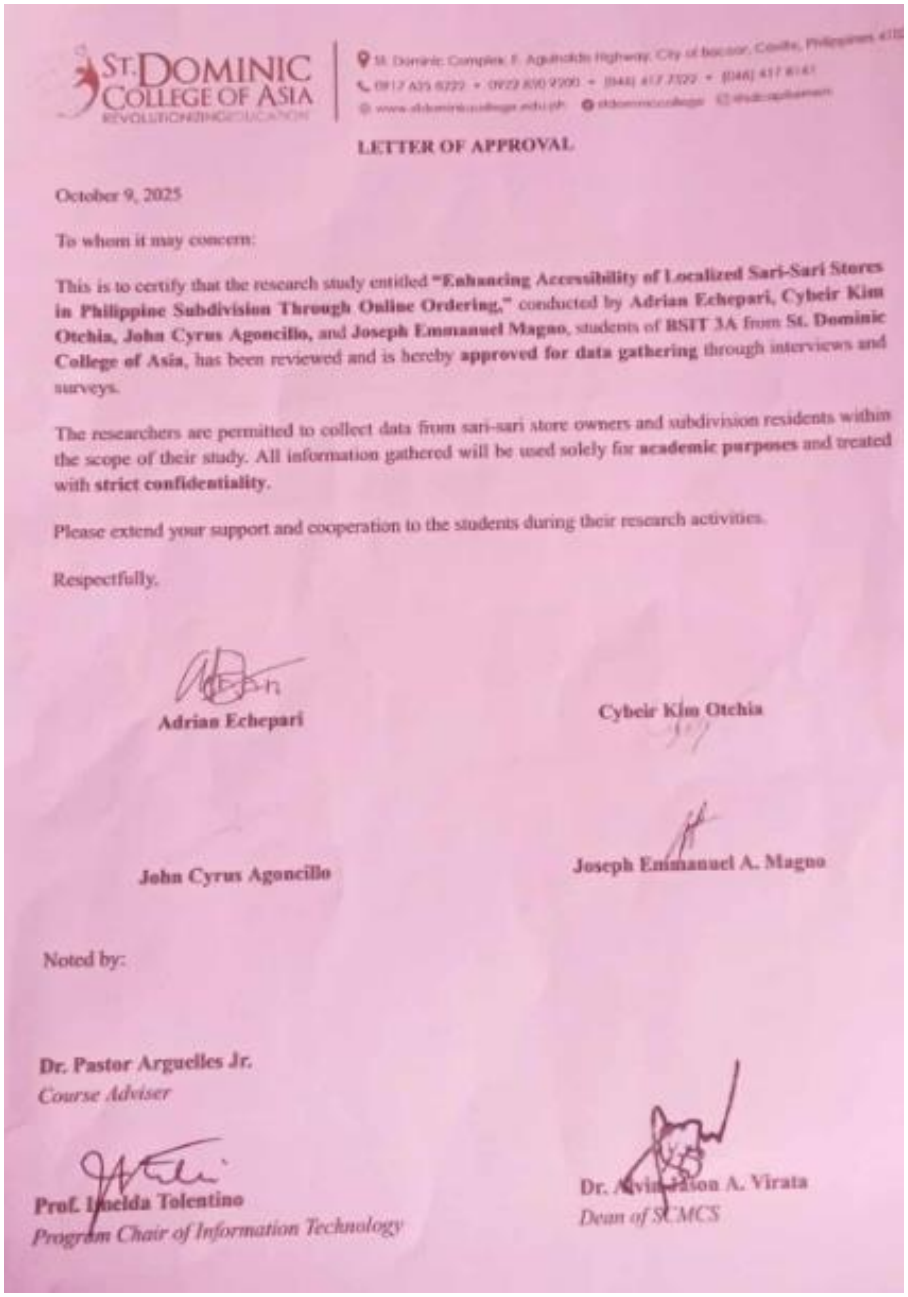
Actions:

1. **Git Revert:** Rollback backend/frontend code to last stable commit.
2. **Database Recovery:** Restore MongoDB Atlas snapshot for products, users, and orders.
3. **Frontend Recovery:** Redeploy previous stable build if deployment fails.
4. **Validation Testing:** Confirm that all core functionalities—login, product catalog, inventory updates, order flow—are working correctly.
5. **Documentation:** Maintain deployment instructions, API routes, and version history in GitHub for traceability.

Appendices

Appendix A

Project Approval Form



ST. DOMINIC COLLEGE OF ASIA
REVOLUTIONIZINGEDUCATION

LETTER OF APPROVAL

October 9, 2025

To whom it may concern:

This is to certify that the research study entitled **"Enhancing Accessibility of Localized Sari-Sari Stores in Philippine Subdivision Through Online Ordering,"** conducted by **Adrian Echepari, Cybeir Kim Otchia, John Cyrus Agoncillo,** and **Joseph Emmanuel Magno,** students of **BSIT 3A** from **St. Dominic College of Asia,** has been reviewed and is hereby **approved for data gathering** through interviews and surveys.

The researchers are permitted to collect data from sari-sari store owners and subdivision residents within the scope of their study. All information gathered will be used solely for **academic purposes** and treated with **strict confidentiality.**

Please extend your support and cooperation to the students during their research activities.

Respectfully,

Adrian Echepari

Cybeir Kim Otchia

John Cyrus Agoncillo

Joseph Emmanuel A. Magno

Noted by:

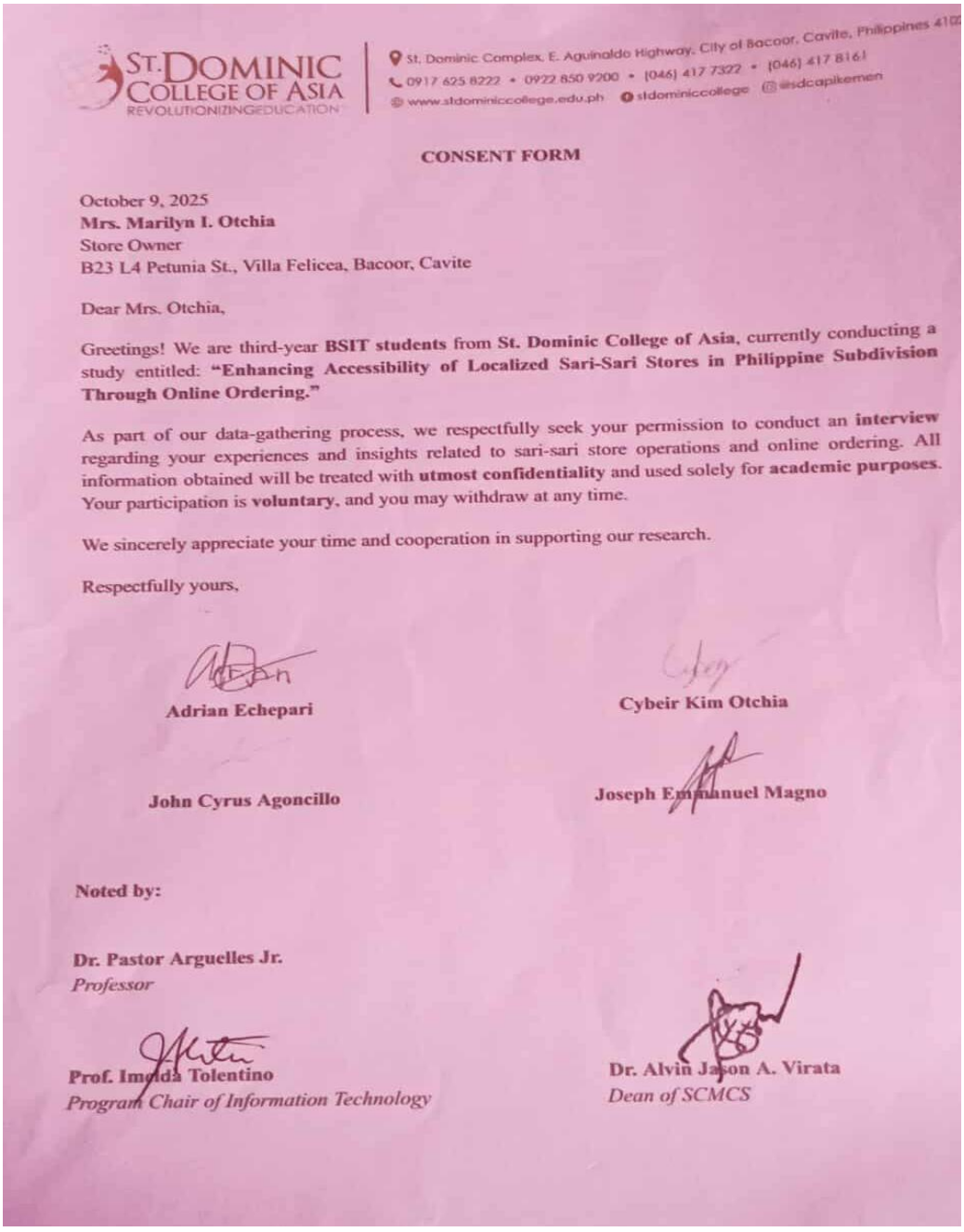
Dr. Pastor Arguelles Jr.
Course Adviser

Prof. Imelda Tolentino
Program Chair of Information Technology

Dr. Alvin Jason A. Virata
Dean of SCMCS

Appendix B

Project Consent Form



ST. DOMINIC COLLEGE OF ASIA
REVOLUTIONIZINGEDUCATION™

St. Dominic Complex, E. Aguinaldo Highway, City of Bacoor, Cavite, Philippines 4102
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CONSENT FORM

October 9, 2025
Mrs. Marilyn I. Otchia
Store Owner
B23 L4 Petunia St., Villa Felicea, Bacoor, Cavite

Dear Mrs. Otchia,

Greetings! We are third-year **BSIT students** from **St. Dominic College of Asia**, currently conducting a study entitled: **"Enhancing Accessibility of Localized Sari-Sari Stores in Philippine Subdivision Through Online Ordering."**

As part of our data-gathering process, we respectfully seek your permission to conduct an **interview** regarding your experiences and insights related to sari-sari store operations and online ordering. All information obtained will be treated with **utmost confidentiality** and used solely for **academic purposes**. Your participation is **voluntary**, and you may withdraw at any time.

We sincerely appreciate your time and cooperation in supporting our research.

Respectfully yours,

Adrian Echepari

Cybeir Kim Otchia

John Cyrus Agoncillo

Joseph Emmanuel Magno

Noted by:

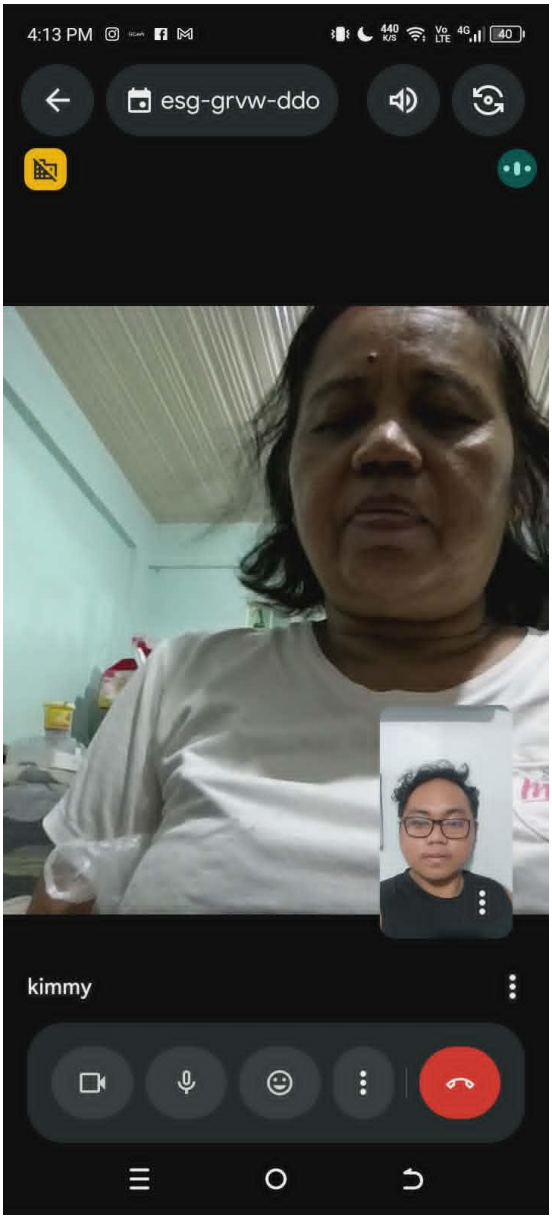
Dr. Pastor Arguelles Jr.
Professor

Prof. Imelda Tolentino
Program Chair of Information Technology

Dr. Alvin Jason A. Virata
Dean of SCMCS

Appendix C

Interview Photo Documentation



Appendix D

Transcript of Interview

Interviewer: Cybeir Kim Otchia

Interviewee: Marilyn Otchia

Date of Interview: October 10, 2025

Location: Google Meet

Duration: 19 Minutes

[00:00–00:30]

Cybeir Otchia: Good afternoon, ma'am. Thank you for joining this interview for the SariSite project. First of all, please introduce yourself and your business.

Marilyn Otchia: Good afternoon, Cybeir. I'm Marilyn Otchia and my business is that I have owned and runned a sari-sari store for 10 years now.

[0:30–0:45]

Cybeir Otchia: Besides you, who usually manages the store?

Marilyn Otchia: Mostly me, but sometimes a family member helps when I'm busy.

[00:45–1:01]

Cybeir Otchia: How many people help you run it?

Marilyn Otchia: Usually one or two, depending on who's available.

[1:02–1:50]

Cybeir Otchia: Okay, so what about the operating hours of your store?

Marilyn Otchia: We're open from morning to evening, everyday.

Cybeir Otchia: May I know the exact time?

Marilyn Otchia: Usually 6:00 AM to 9:00 PM, pero minsan lumalagpas pa doon depende sa kapag may bumibili pa.

[1:51–2:38]

Cybeir Otchia: What products do you mainly sell?

Marilyn Otchia: Yung usual na binibili sa mga sari-sari store — snacks, canned goods, condiments, hygiene products, saka ano- yung mga sangkap rin tulad ng itlog, kamatis, ganon.

[2:58–05:10]

Cybeir Otchia: Can you describe your step-by-step process from receiving an order to completing a sale?

Marilyn Otchia: So una, syempre tatawagin ako, tapos sasabihin nung customer lahat ng mga bibilhin niya, kukunin ko pa isa-isa lahat, tapos bibigay sakin yung bayad, aabutan ko ng sukli kapag meron- ay ano rin pala, icacalcu ko muna kung magkano lahat tapos pati narin yung sukli, sinasama ko sa pag compute. Tapos isusulat ko usually sa notebook yung mga binili saka kita.

[05:10–06:00]

Cybeir Otchia: What do you do when an item is out of stock?

Marilyn Otchia: Chinecheck ko muna kung meron pa kami non sa likod pero kapag wala talaga, sinasabi ko sa kanila o kaya ano- kungyari, bumibili sila ng lorin's patis pero wala, tatanong ko sa kanila kung okay lang Datu Puti patis, ganon.

[06:00–07:45]

Cybeir Otchia: What usually causes delays in your store operations?

Marilyn Otchia: Tulad nung sinabi ko kanina, medyo mahaba yung proseso nung lahat, lalo na kung madami yung mga nabili tapos madami rin yung binibili, parang grocery lng haha. Kaya madalas talaga, kailangan ko ng tulong.

[07:45–08:28]

Cybeir Otchia: Do you have store policies for cancellations, returns, or replacements?

Marilyn Otchia: Yes. Pwede naman kapag kungware kilala ko naman yung bumibili, nagkamali lang yung nabili, baka inutusan lang ng nanay ganon, syempre naman, ibabalik ko yung bayad niya basta ibalik rin yung binili.

[08:28–09:50]

Cybeir Otchia: Do you follow any rules for pricing, discounts, or bundles?

Marilyn Otchia: Nilalagyan ko lang ng tubo yung SRP talaga, saka hindi ako nag bubundles or discount.

[09:50–10:48]

Cybeir Otchia: How often do you conduct inventory checks?

Marilyn Otchia: Usually after every night kada magsasara na kami kaso minsan nakakalimutan ko kaya pag dating ng kinabukasan, don ko lang mapapansin na kulang na pala yung stock kasi may bumili na customer tapos ano- pagkatingin ko, wala na pala kami non.

[10:48–11:30]

Cybeir Otchia: Do you provide delivery services now?

Marilyn Otchia: Minsan, kapag kakilala namin, tinetext lang ako or chat sa messenger tapos may inuutusan ako para mga deliver don sa kanila kasi alam naman namin kung san sila nakatira.

[11:30–12:18]

Cybeir Otchia: What daily tasks would you like SariSite to automate for you?

Marilyn Otchia: Maganda sana kung less hassle, alam ko na kagad lahat ng bibilhin nila bago pa sila pumunta dito para pagdating nila, bayad nalang saka bigay. Okay rin sana kung hindi na kailangan magsulat sa notebook, yung ano- automatic kumbaga- yung pagkaano- track ng mga stock namin.

[12:18–13:05]

Cybeir Otchia: What notifications do you want to receive in regards to your store?

Marilyn Otchia: Hmm.. siguro kung may ano- may alam kong padating na bibili, saka kapag paubos na yung stock ko. Pwede rin ano- syempre kapag may nagchachat sakin ganon na bibili or mag o-order.

[13:05–13:45]

Cybeir Otchia: Kung pwede sa cellphone, mas prefer niyo po ba yon para gamitin yung system na ide-develop namin?

Marilyn Otchia: Ay syempre, haha. Lagi malapit sakin ang cellphone ko, lalo na kapag wala masyadong nabili, naghihintay lang ako doon, yun na rin nagiging libangan ko.

[13:45–14:25]

Cybeir Otchia: Would you prefer a simple interface or a system with advanced analytics?

Marilyn Otchia: Ay hindi ako maalam diyan sa ano- technology. Mas gugustuhin ko kapag madali lang gamitin.

[14:25–15:00]

Cybeir Otchia: Do you have stable internet and a device to access the system?

Marilyn Otchia: Yes, meron naman pareho.

[15:00–16:00]

Cybeir Otchia: Do you have any concerns about data privacy and security?

Marilyn Otchia: Wala naman masyado, basta lang naman syempre wag kumalat yung mga cellphone number ng mga customer ko pati address nila, lalo na ngayon, madaming kumakalat or ano- nagsisilabasan pala na mga scammer, delikado na.

[16:00–16:48]

Cybeir Otchia: Would you need orientation or training to use SariSite?

Marilyn Otchia: Kahit video tutorial lang sana or kahit turuan mo nalang ako kung paano gamitin pala, mas okay, haha.

[16:48–17:40]

Cybeir Otchia: How often do you need sales or inventory reports?

Marilyn Otchia: As long as kumikita kami kada linggo, okay na siya.

[17:40–18:30]

Cybeir Otchia: Are there other tools or features you think would be helpful?

Marilyn Otchia: Hmm… wala naman na yata maliban doon sa mga nasabi ko na kanina—automatic yung proseso saka yung ano- yung iba rin.

[18:30–19:00]

Cybeir Otchia: Thank you for your time, ms. Marilyn. This information will definitely be helpful to us in designing a system na sure kaming magugustuhan niyo and makakatulong sa Sari-Sari store niyo. Maraming salamat po ulit.

Marilyn Otchia: Thank you as well, Cybeir. Excited akong makita yung “SariSite”, pag tapos na.

Signed by



Marilyn Otchia

Interviewee

Appendix E

Google Forms Survey for Other Sari-Sari Store Owners

Survey on Enhancing Accessibility of Localized Sari-Sari Stores Through Online Ordering (**Store Owner** Questionnaire)

We are BSIT students from St. Dominic College of Asia conducting a research study about improving the accessibility of sari-sari stores in Philippine subdivisions through an online ordering system. This survey aims to gather your insights and experiences. Your answers will be kept confidential and used for academic purposes only.

This form is automatically collecting emails from all respondents. Change settings

Full Name *

Short answer text

Email Address *

Short answer text

Data Privacy Act of 2012

In accordance with the RA 10173 or Data Privacy Act of 2012, the information you will provide will be treated with utmost confidentiality. No other personal information allowing an identification will be asked from you as a participant.

Moreover:

1. I acknowledge that the researcher has clearly explained to me what the survey/questionnaire and the study is all about;
2. I understand that the survey/questionnaire will not directly benefit me or that I understand potential benefits and/or risks;
3. I agree for my responses be solely used for this research alone;
4. I, the participant agrees to the duration of the time of the survey;
5. I may withdraw from the survey/data collection procedure without consequences;
6. that the data will be treated with confidentiality;
7. that I will remain anonymous; and
8. that the I may contact the researcher for further clarification and information.

Do you consent to answering this questionnaire? *

- ○ Yes, I give my consent to answering this questionnaire
- ○ No, I do not give my consent

Section 2 of 2

Questions for a Sari-Sari Store Owner Regarding the Adoption of an Online System

Description (optional)

How many years have you been managing your Sari-sari store? *

Short answer text

What challenges do you encounter in managing daily transactions? *

Long answer text

How do you currently monitor and record your inventory and sales? *

Long answer text

What difficulties do you experience when handling customer orders and deliveries? *

Long answer text

What features would you find helpful in an online sari-sari store management system? *

Long answer text

What potential challenges do you anticipate in adopting an online system for your store? *

Long answer text

Questions for a Sari-Sari Store Owner Regarding the Adoption of an Online System

How many years have you been managing your Sari-sari store?

4 responses

- 10years
- 5years
- 6 years
- 10

What challenges do you encounter in managing daily transactions?

4 responses

- Some customers don't know how to pay their debts properly — they pay using damaged or even fake money.
- forgetting customer utang, counting stocks accurately.
- Minsan panukli nauubusan ng barya
- The main challenges are tracking sales and inventory, handling "utang," managing small change, and keeping up with price changes.

How do you currently monitor and record your inventory and sales?

4 responses

- I record sales every day.
- I currently record my inventory and sales manually using a notebook
- Chinecheck isa isa
- I usually record sales and inventory manually in a notebook, and sometimes I just check the stock by counting items on the shelves.

What difficulties do you experience when handling customer orders and deliveries?

4 responses

Some customers are not honest, and their orders are sometimes delivered late.

I sometimes experience confusion or mix-ups with orders.

Pag madami yung order at pag matagal yung rider

Sometimes orders get mixed up or delayed, especially when there are many customers.

What features would you find helpful in an online sari-sari store management system?

4 responses

Inventory tracking (to monitor stocks and get low-stock alerts)

Sales and expense records (to see profit and losses)

Utang tracking for customers

Mobile-friendly and easy to use interface

Payment options (GCash, PayMaya, cash)

Reports and analytics for best-selling items

Supplier and restock reminders

Offline access in case there's no internet.

inventory tracking, sales and expense recording, utang management, low-stock alerts.

Mas mapapadali ang pag kuha ng mga orders

What potential challenges do you anticipate in adopting an online system for your store?

4 responses

I might face challenges like poor internet connection.

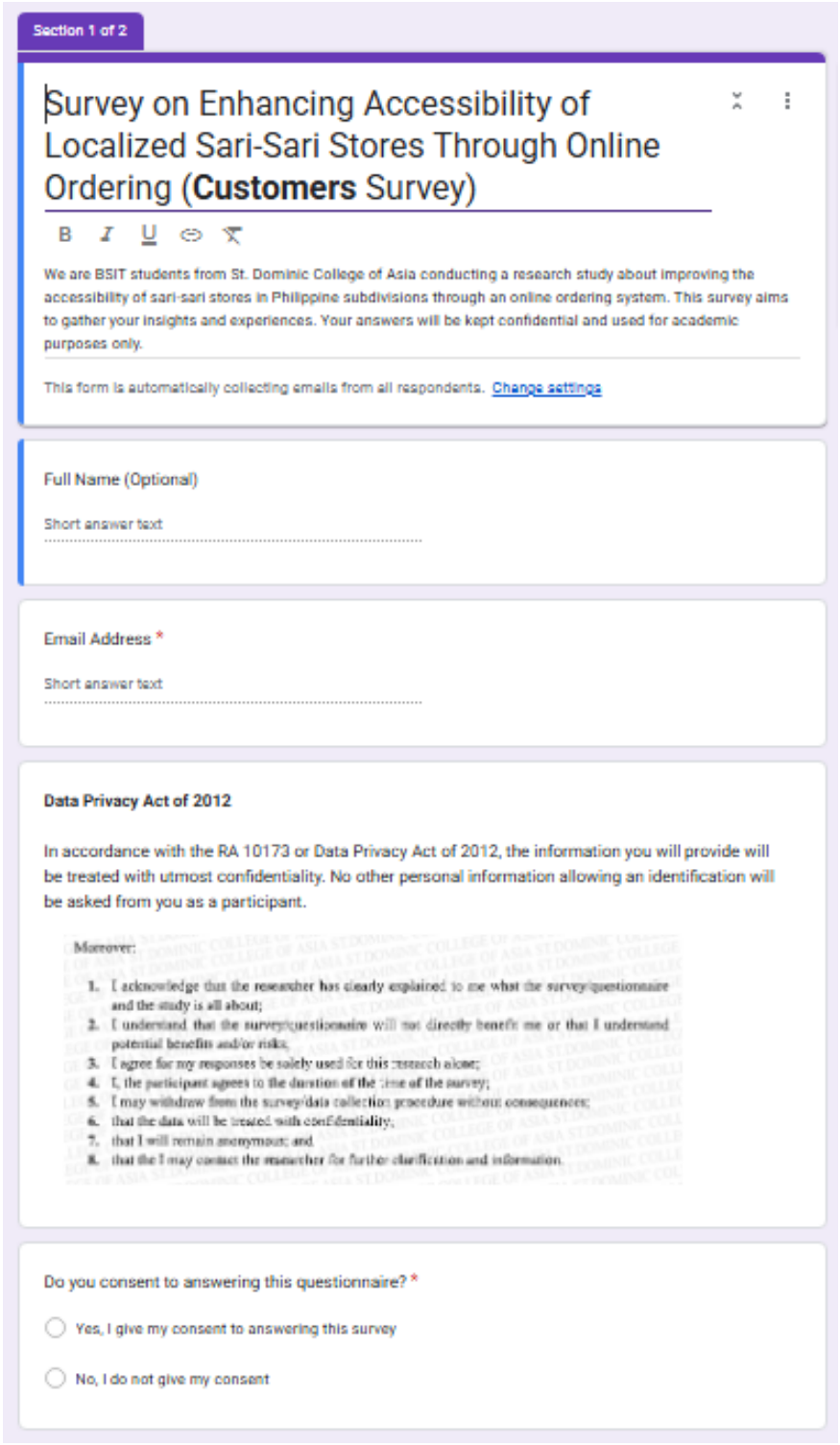
I anticipate are slow or unstable internet connection, difficulty learning the new system, possible technical errors.

Siguro pag nawalan ng net or pag nagloko yung net

Some challenges might be learning how to use the system, needing a stable internet connection, and the risk of technical issues or data loss.

Appendix F

Google Form Survey for Customers of Sari-sari Stores



Section 1 of 2

Survey on Enhancing Accessibility of Localized Sari-Sari Stores Through Online Ordering (**Customers** Survey)

We are BSIT students from St. Dominic College of Asia conducting a research study about improving the accessibility of sari-sari stores in Philippine subdivisions through an online ordering system. This survey aims to gather your insights and experiences. Your answers will be kept confidential and used for academic purposes only.

This form is automatically collecting emails from all respondents. Change settings

Full Name (Optional)

Short answer text

Email Address *

Short answer text

Data Privacy Act of 2012

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Moreover:

1. I acknowledge that the researcher has clearly explained to me what the survey/questionnaire and the study is all about;
2. I understand that the survey/questionnaire will not directly benefit me or that I understand potential benefits and/or risks;
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6. that the data will be treated with confidentiality;
7. that I will remain anonymous; and
8. that the I may contact the researcher for further clarification and information.

Do you consent to answering this questionnaire? *

- Yes, I give my consent to answering this survey
- No, I do not give my consent

Section 2 of 2

Questions for Customers Regarding Sari-Sari Stores

Description (optional)

Do you often purchase from sari-sari stores in your subdivision? Why or why not? *

Long answer text

Would you agree that ordering from a sari-sari store online instead of visiting physically would be more convenient? *

	1	2	3	4	
Strongly Disagree	○	○	○	○	Strongly Agree

If delivery service were available, would you be more likely to use it when buying daily necessities? *

	1	2	3	4	
Strongly Disagree	○	○	○	○	Strongly Agree

Do you believe that having an online ordering and delivery system will improve your shopping experience with sari-sari stores? *

	1	2	3	4	
Strongly Disagree	○	○	○	○	Strongly Agree

Would seeing the available stock of items online make it more convenient for you to shop at a sari-sari store? *

	1	2	3	4	
Strongly Disagree	○	○	○	○	Strongly Agree

Would you be confident in using a website or app for ordering from a sari-sari store? *

	1	2	3	4	
Strongly Disagree	○	○	○	○	Strongly Agree

Is it important for you to receive notifications (e.g., order confirmation, delivery status, item restocked) from a sari-sari store's online system? *

	1	2	3	4	
Strongly Disagree	○	○	○	○	Strongly Agree

I would find a pre-order or reservation feature useful for items that run out quickly. *

	1	2	3	4	
Strongly Disagree	○	○	○	○	Strongly Agree

Would you recommend an online sari-sari store to your neighbors or family if it is easy to use and makes buying from sari-sari stores more convenient? *

	1	2	3	4	
Strongly Disagree	○	○	○	○	Strongly Agree

Questions for Customers Regarding Sari-Sari Stores

Do you often purchase from sari-sari stores in your subdivision? Why or why not?

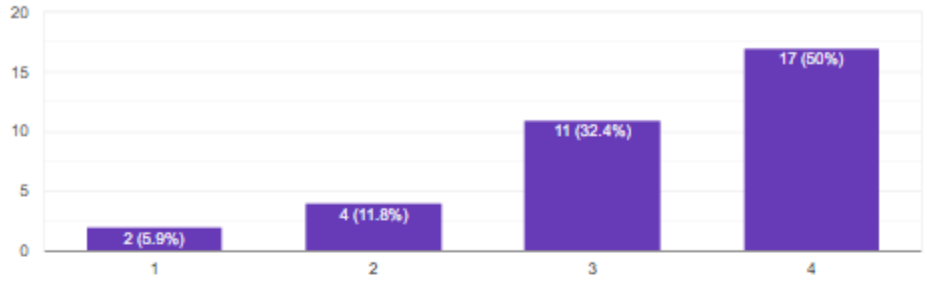
34 responses

- yes its more convenient
- Yes for my everyday needs
- Yes, sari-sari stores are convenient for me whenever I need something immediately.
- Yes, because it's more accessible rather than going to mall/supermarket
- Yes, because it's close to our house
- Sometimes. Because they closer and convenient.
- yes, because we actually own one
- Yes for daily goods i need in a daily basis.
- Yes. It's near my home and it's cheap.

Would you agree that ordering from a sari-sari store online instead of visiting physically would be more convenient?

Copy chart

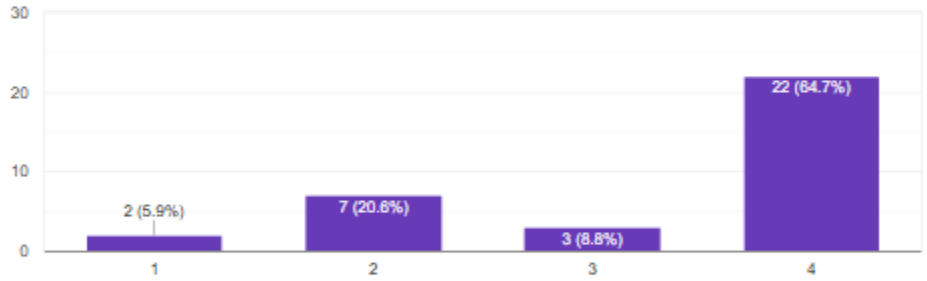
34 responses



If delivery service were available, would you be more likely to use it when buying daily necessities?

Copy chart

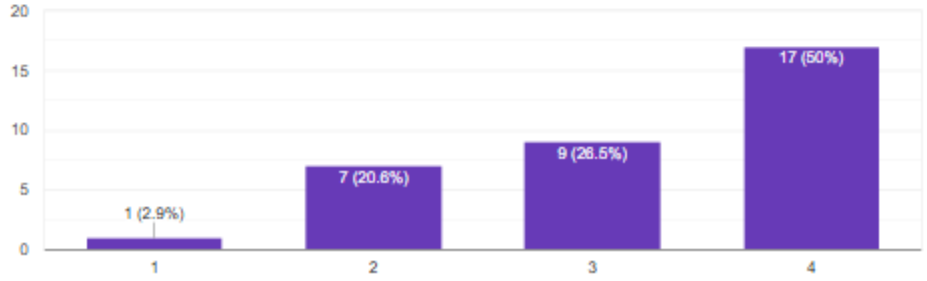
34 responses



Do you believe that having an online ordering and delivery system will improve your shopping experience with sari-sari stores?

Copy chart

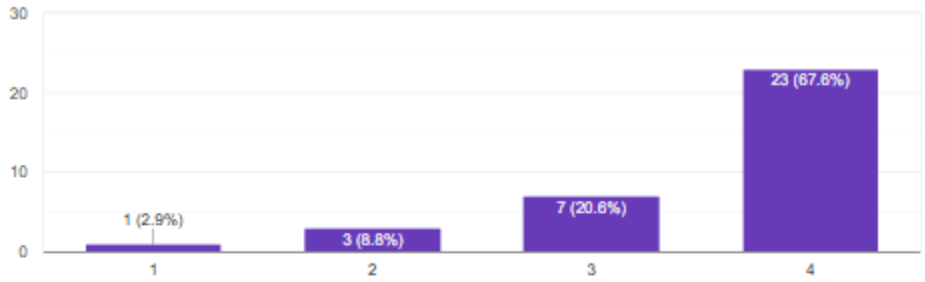
34 responses



Would seeing the available stock of items online make it more convenient for you to shop at a sari-sari store?

Copy chart

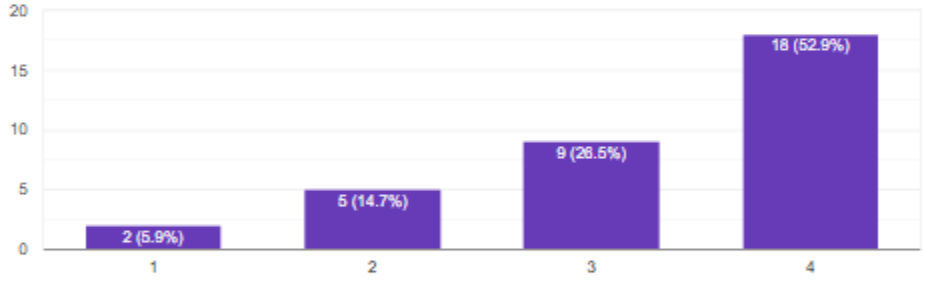
34 responses



Would you be confident in using a website or app for ordering from a sari-sari store?

Copy chart

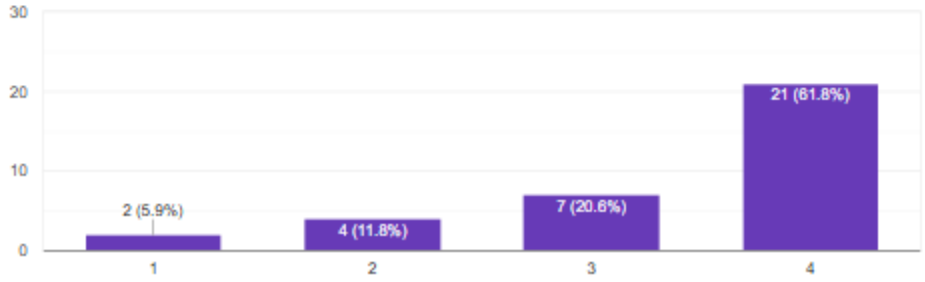
34 responses



Is it important for you to receive notifications (e.g., order confirmation, delivery status, item restocked) from a sari-sari store's online system?

Copy chart

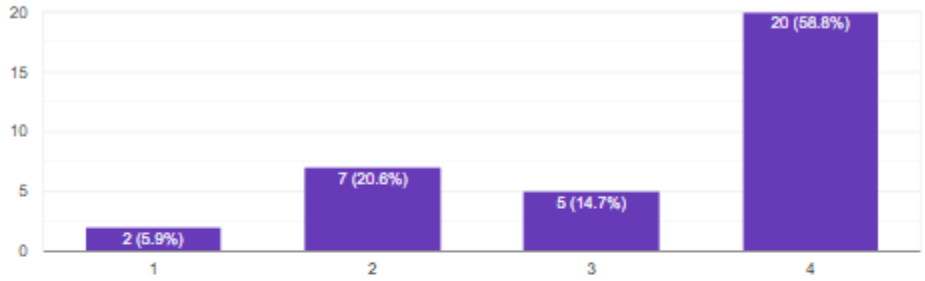
34 responses



I would find a pre-order or reservation feature useful for items that run out quickly.

Copy chart

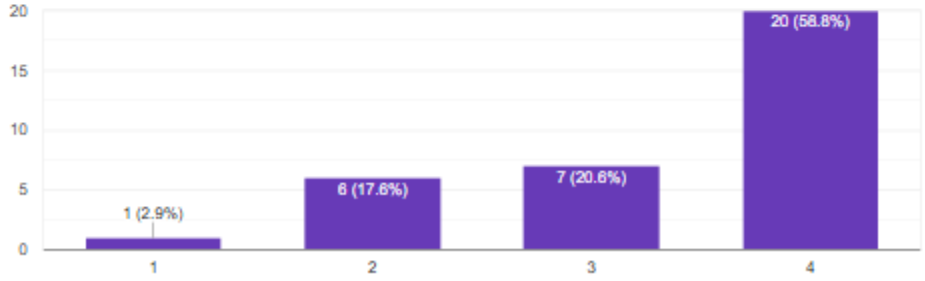
34 responses



Would you recommend an online sari-sari store to your neighbors or family if it is easy to use and makes buying from sari-sari stores more convenient?

Copy chart

34 responses



Appendix E

Curriculum Vitae of the Researchers

John Cyrus Agoncillo

Software developer



PROFILE

Inspired and driven college graduate seeking an entry-level Software Developer position at a forward-thinking organization. Brings a strong academic foundation in programming, data analysis, and project collaboration, with a passion for learning, problem-solving, and contributing to team success

SKILLS

- Process improvement
- Fast Learner
- Coding Skills
- Hard worker
- Critical thinking skills
- Computer Skills
- Proactive and self-motivated
- Understanding of data structures and algorithms

EDUCATION

Elementary - Senior High School

Our Lady of The Pillar Catholic School inc — **SY 2010 - 2023**

College

St. Dominic College of Asia - Bachelor of Science in Information Technology. — **SY 2023**

CERTIFICATIONS

Microsoft Office Specialist: Word Associate (Office 2019)
Date issued: March 13, 2024

IT Specialist - Databases
Date issued: May 06, 2025

CONTACT

Contact Number: 09563183193
Email: PhmYT0403@gmail.com
Address: Bellefort Estate, Blk 9 , Lot 4 - Molino 4 Bacoor City, Cavite



Adrian Echepari

Software Developer

About Me

I know how to solve problems in the most efficient way. I am excellent at managing my time, and I am also a very creative person when it comes to all of my projects. If I want to do something, I'll find a way.

Contact

+63 9273725242

adrian.echepar07@gmail.com

Bacoor, Cavite, Philippines

Skills

- Web Design
- Touch Typing
- Software Development
- Fast Learner
- Video Editing

Language

- English
- Tagalog

Education

(2011 - 2017)
RUTHER E. ESCONDE'S SCHOOL OF MULTIPLE INTELLIGENCES INC.
Elementary School

(2017 - 2021)
MILLENIUM CHRISTIAN HIGH SCHOOL OF CAVITE
High School

(2021 - 2023)
ST. DOMINIC COLLEGE OF ASIA
Senior High School
TVL-ICT

(2023 - Present)
ST. DOMINIC COLLEGE OF ASIA
College
BS in Information Technology

Joseph Emmanuel A. Magno

Software developer



PROFILE

Motivated and results-driven College graduate seeking a position within a large organization as a Software Developer. Offering a strong foundation in coding skills, data analysis, and project management, with a proven ability to drive efficiency, deliver successful outcomes and collaborate within cross-functional teams.

SKILLS

Process improvement	Hard worker	Computer Skills
Data-driven strategic planning	Critical thinking skills	Proactive and self-motivated
Fast Learner	Coding Skills	

EDUCATION

Elementary

Dulong Bayan Elementary School — **SY 2010 - 2015**

Junior High School

Waynesville Learning Center — **SY 2016 - 2019**

Senior High School

University of Perpetual Help Molino Campus - STEM strand. — **SY 2020 - 2022**

College

St. Dominic College of Asia - Bachelor of Science in Information Technology. — **SY 2021 - 2022**

National University Dasmariñas - Bachelor of Science in Information Technology with specialization in Mobile and Web Applications. — **SY 2023 - 2024**

St. Dominic College of Asia - Bachelor of Science in Information Technology. — **SY 2024 - 2025**

CONTACT

Contact Number: 09272887883
Email: josephemmanuelmagno@gmail.com
Address: Blk 5 lot 7 Phs 1 Veraville Arcadia Dulong Bayan Bacoor City Cavite



Cybeir kim Otchia

College Student

Summary

Motivated and detail-oriented BS in Information Technology student with a strong foundation in software development, networking, and database management. Eager to apply academic knowledge to real-world projects through internships or entry-level opportunities. Demonstrates strong problem-solving skills, adaptability, and a willingness to learn new technologies. Passionate about building innovative solutions and contributing to team success in dynamic tech environments.

Education

2023 – CurrentBachelor of Science in Information Technology St. Dominic College of Asia – Bacoor, CAV

Technical skills

- Programming
- Web Design
- Database Management
- Software
- Microsoft Office

Soft skills

- Communication skills
- Problem-solving
- Willingness to learn
- Team building
- Deadline management

Certifications

- Information Technology Specialist – Databases

Issued by Certiport | Pearson VUE – May 7, 2025
Credential ID: mq4L-uTnH
verify.certiport.com

- Microsoft Office Specialist – PowerPoint 2019

Issued by Microsoft – May 30, 2024
Credential ID: vmGBN-48Xu
verify.certiport.com

Language

- **English**
- **Filipino**

Contact

- **Bacoor Cavite, Philippines**
- **09127517390**
- **otchiakimn@gmail.com**

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Chilton, M. A. (2006). Data Modeling Using Entity Relationship Diagrams: A Step-Wise Method. Journal of Information Systems Education, 17(4), 385–394.

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Resource Persons

Name: Ms. Marilyn Otchia
Role: Sari-sari Store Owner / Main Stakeholder
Affiliation: Alyn's Sari-sari Store
Date(s) Consulted: October 10, 2025
Contribution: Provided an in-depth interview about the current operations of Alyn's Sari-sari Store, including product listing practices, inventory handling, order processes, and payment workflows. She also explained the store's standard procedures and daily operational routines, which served as the basis for understanding real business requirements.

Name: Sir Sylvan Cahilog
Role: Professor
Affiliation: St. Dominic College of Asia
Date(s) Consulted: September 20,2025 | October 4,2025 | October 25, 2025 | November 8, 2025 | November 15,2025 | November 22,2025
Contribution: Gave professional feedback on the system's design and functionality, offering suggestions on what to enhance, revise, and troubleshoot. He also guided the research direction by providing academic tips, corrections, and advice to improve the overall quality of the research paper.

Name: Dr. Pastor Arguelles Jr.
Role: Professor / Research Adviser
Affiliation: St. Dominic College of Asia
Date(s) Consulted: August 19, 2025 | October 6, 2025 | September 26, 2025 | November 3, 2025
Contribution: Provided expert guidance throughout the research process by reviewing the system's design and functionality, suggesting improvements, and identifying issues to troubleshoot. Offered academic advice, corrections, and strategies to enhance the research methodology and improve the overall quality and rigor of the research paper.